

## RESEARCH ARTICLE

# DNA Binding by BosR Controls RpoS-Dependent and -Independent Gene Expression in *Borrelia burgdorferi*

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## ABSTRACT

BosR, the sole member of the ferric uptake regulator (FUR) family in *Borrelia burgdorferi*, is essential for the spirochete's transcriptional adaptation to the mammalian host environment. Although best known for activating *rpoS* and establishing the mammalian-phase RpoS regulon, BosR originally was linked to regulation of genes involved in *B. burgdorferi*'s oxidative stress response. Here, we show that BosR governs gene expression through both RpoS-dependent and RpoS-independent mechanisms under in vitro and mammalian host-adapted conditions. Using RNA-seq and a DNA-binding-defective BosR-R39A mutant, we demonstrate that DNA binding is essential for BosR's global regulatory functions. BosR activates *rpoS*, promotes RpoS-dependent gene regulation, and independently modulates a distinct set of genes involved in a variety of cellular functions, including genome maintenance, chemotaxis, and virulence. Notably, canonical oxidative stress response genes previously attributed to BosR were not differentially expressed in  $\Delta$ *bosR* strains in vitro or in mammals. Despite its broad regulatory scope, BosR does not recognize a single, conserved DNA-binding motif, suggesting that DNA occupancy is influenced by local sequence context or DNA topology. Our findings support a bifunctional model in which BosR collaborates with RNA polymerase (RNAP)-RpoS holoenzyme to activate and repress RpoS-regulated genes, while functioning in a FUR-like manner to control RpoD-dependent genes independently of RNAP interaction.

## 1 | Introduction

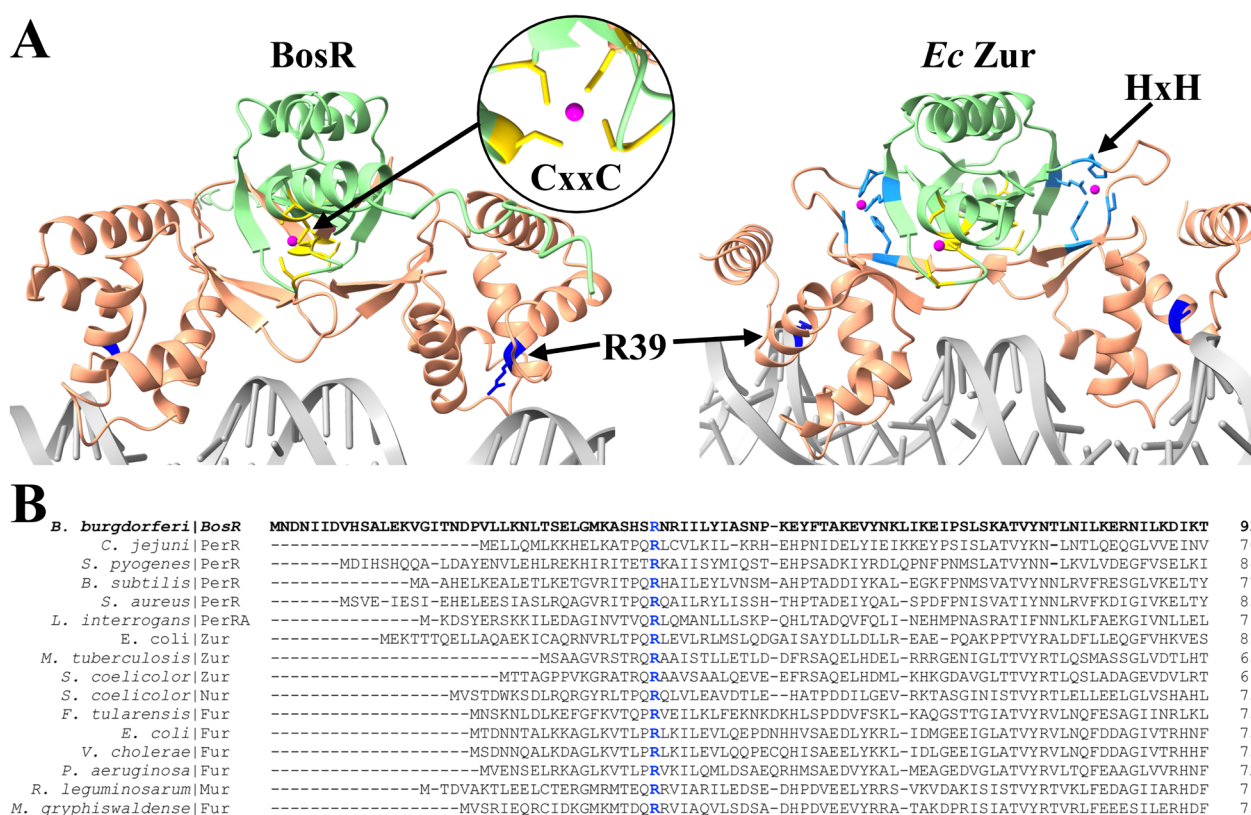
Lyme disease (LD) is an inflammatory, arthropod-borne illness caused by infection with the spirochete *Borrelia burgdorferi* (Radolf et al. 2012). In nature, *B. burgdorferi* cycles between an *Ixodes* spp. tick vector and a vertebrate reservoir host, typically small rodents (Telford and Goethert 2020). Humans, incidental, dead-end hosts, become infected when they intrude upon this enzootic cycle and provide a blood meal for an infected tick (Radolf et al. 2012). Human encounters

with ticks are frequent enough that LD is now the most prevalent arthropod-borne disease in the United States, Europe and Asia (Kugeler et al. 2024; Davidson et al. 2025; Song et al. 2025). In the United States alone, an estimated 476,000 cases of LD occur annually (Kugeler et al. 2021). To sustain its dual-host lifestyle, *B. burgdorferi* relies on sophisticated regulatory mechanisms to reshape its transcriptome in response to tick- and mammalian host-derived signals. Although numerous studies over the years have provided insight into *B. burgdorferi*'s parsimonious gene regulatory pathways and

their roles in LD pathogenesis (Radolf et al. 2012; Samuels et al. 2021; Stevenson 2023; Stevenson et al. 2022), the interplay between them remains poorly understood.

The compact genome of *B. burgdorferi* encodes only a handful of identifiable transcriptional regulators (Samuels et al. 2021). Among the best characterized is BosR (*Borrelia* oxidative stress regulator), *B. burgdorferi*'s sole member of the FUR (ferric uptake regulator) family (Boylan et al. 2003; Seshu et al. 2004; Hyde et al. 2009; Ouyang et al. 2009; Hyde et al. 2010; Ouyang et al. 2011; Wang et al. 2013; Shi et al. 2014; Katona 2015; Ouyang et al. 2015; Mason et al. 2019; Grassmann et al. 2023; Raghunandan et al. 2024, 2025). While BosR is structurally similar to canonical FURs (Grassmann et al. 2023; Mason et al. 2019) (Figure 1A), it lacks the hallmark regulatory metal binding site—a conserved His/Glu/Asp cluster required for coordinating divalent regulatory metals—found in nearly all other FUR orthologs (Steingard and Helmann 2023; Sevilla et al. 2021; Sarvan, Chari, et al. 2018; Sarvan, Butcher, et al. 2018; Liu et al. 2021; Lee and Helmann 2007; Kang

et al. 2024; Gilston et al. 2014; Deng et al. 2015). However, it does retain the CxxC zinc-binding motif necessary for dimerization and activity (Mason et al. 2019), as well as conserved residues in its helix-turn-helix DNA-binding domain, including arginine 39 (R39) (Katona 2015; Seshu et al. 2004), a DNA-contacting residue in other FUR proteins (Sarvan, Butcher, et al. 2018; Sarvan, Chari, et al. 2018; Sevilla et al. 2021) (Figure 1B). BosR initially was described for its putative role regulating genes involved in *B. burgdorferi*'s oxidative stress response, including *napA/dps* (neutrophil-activating protein A/DNA-binding protein from starved cells) (Hyde et al. 2010; Boylan et al. 2003; Seshu et al. 2004; Hyde et al. 2009), *sodA* (superoxide dismutase) (Hyde et al. 2006, 2010; Seshu et al. 2004; Esteve-Gassent et al. 2009), and *cdr* (coenzyme A disulfide reductase) (Hyde et al. 2009). However, different studies have reported inconsistencies regarding which oxidative stress response genes, if any, are regulated by BosR (Hyde et al. 2009; Ouyang et al. 2009, 2011). BosR subsequently gained prominence following the discovery that it binds to the *rpoS* promoter and is essential for RpoN-dependent



**FIGURE 1** | Sequence and structural analysis of BosR and canonical FUR proteins. (A) Comparative analysis of the AlphaFold3-predicted structure of BosR (left) and the crystal structure of *E. coli* Zur (Zinc uptake regulator; PDB ID 4MTD; right), a canonical FUR family regulator. Both proteins contain a C-terminal dimerization domain (green) with a conserved CxxC motif (yellow) that coordinates zinc (purple) to promote dimerization and an N-terminal helix-turn-helix DNA-binding domain (salmon) that interacts with DNA via residues that include a conserved arginine at position 39 in BosR (blue). BosR lacks the regulatory metal-binding site (HxH motif, cyan) found in Zur and other canonical FUR proteins. (B) Multiple sequence alignment (MSA) of the DNA-binding domain of BosR and representative FUR proteins shows conservation of key residues, including arginine 39 (blue). UniProt and PDB IDs for FUR proteins included in the MSA: *Campylobacter jejuni* PerR (Q0PBI7; PDB: 6DK4); *Streptococcus pyogenes* PerR (A0A0H2UT39; PDB: 4I7H); *Bacillus subtilis* PerR (P71086; PDB: 3F8N); *Staphylococcus aureus* PerR (Q2G282); *Leptospira interrogans* PerRA (Q72QS5; PDB: 5NL9); *Escherichia coli* Zur (P0AC51, PDB: 4MTD) and Fur (P0A9A9, PDB: 2FU4); *Mycobacterium tuberculosis* Zur (P9WN85, PDB: 2O03); *Streptomyces coelicolor* Zur (Q9L2H5, PDB: 3MWM) and Nur (Q9K4F8, PDB: 3EYY); *Francisella tularensis* Fur (Q5NIN6, PDB: 5NBC); *Vibrio cholerae* Fur (P0C6C8; PDB: 2W57); *Pseudomonas aeruginosa* Fur (Q03456, PDB: 6H1C); *Rhizobium leguminosarum* Mur (O07315, PDB: 5FD6); and *Magnetospirillum gryphiswaldense* Fur (V6F4Q0, PDB: 4RB1).

expression of *rpoS* (Hyde et al. 2009, 2010; Ouyang et al. 2011, 2009, 2015; Katona 2015). More recently, Raghunandanan et al. (2024) proposed that BosR enhances RpoS levels by binding to the 5' untranslated region of *rpoS* mRNA to protect it from degradation rather than through direct interaction with the *rpoS* promoter (Raghunandanan et al. 2024). Our recent work has expanded upon BosR's known regulatory functions by showing that, in addition to activating *rpoS*, in mammalian host-adapted spirochetes BosR licenses transcription of RpoS-dependent genes and RpoS-mediated repression of tick-phase genes (Grassmann et al. 2023). These global effects of BosR on the mammalian-phase RpoS regulon suggest that BosR interacts directly with both DNA and the RNA polymerase (RNAP)-RpoS holoenzyme at RpoS-regulated promoters (Grassmann et al. 2023).

For the past two decades, the LD field has focused almost exclusively on BosR's role in the expression of *rpoS* and RpoS-regulated genes (Grassmann et al. 2023; Hyde et al. 2009, 2010; Katona 2015; Mason et al. 2019; Ouyang et al. 2011, 2009, 2015; Raghunandanan et al. 2024; Seshu et al. 2004; Shi et al. 2014; Sze and Li 2023; Wang et al. 2013). BosR's FUR-like regulatory functions outside the RpoN/RpoS pathway were far less studied (Ouyang et al. 2009; Wang et al. 2017). Herein, we investigated FUR-like functions of BosR by comparing the transcriptomes of wild-type (WT) and  $\Delta$ *bosR* *B. burgdorferi* strains cultivated in vitro and in our dialysis membrane chamber (DMC) system for generating mammalian host-adapted spirochetes (Grassmann et al. 2023; Caimano 2018; Akins et al. 1998). This analysis revealed that, in addition to *rpoS* and RpoS-regulated genes (Caimano et al. 2005; Caimano et al. 2019; Grassmann et al. 2023), BosR controls the expression of a distinct set of genes independently of RpoS. Notably, the RpoS-independent genes regulated by BosR in DMCs represent a distinct regulon from that observed under in vitro conditions. Importantly, oxidative stress response genes previously linked to BosR (Boylan et al. 2003; Seshu et al. 2004; Hyde et al. 2006, 2009; Ouyang et al. 2009; Esteve-Gassent et al. 2009; Hyde et al. 2010; Wang et al. 2013) were not dysregulated in the  $\Delta$ *bosR* strain under either condition. To examine the functional importance of DNA binding by BosR, we engineered *B. burgdorferi* strains expressing a BosR mutant with an alanine substitution in R39. This substitution abrogated DNA binding and markedly impaired transcription of *rpoS*, the RpoS-regulon, and RpoS-independent genes. Intriguingly, using bioinformatics, we were unable to identify conserved BosR-binding motifs upstream of most BosR-regulated genes. Our data support a model in which BosR is bifunctional, with both activities involving predominantly non-sequence-specific DNA binding. In one role, BosR acts analogously to other known, albeit structurally unrelated, sigma factor activator proteins (Bouillet et al. 2024; Cartagena et al. 2019; Chen et al. 2021; Gottesman 2019; He et al. 2022; Tabib-Salazar et al. 2013; Vishwakarma and Brodolin 2020; Wu et al. 2023, 2018; Hubin et al. 2017, 2015; Desai et al. 2018; Bao et al. 2012; Xu et al. 2019). In this capacity, it specifically tethers RpoS to RNAP, promoting RpoS-dependent gene activation and repression of tick-phase genes. The second role resembles that of a conventional FUR-like function, regulating RpoD-dependent genes through DNA binding without direct interaction with the RNAP-RpoD holoenzyme.

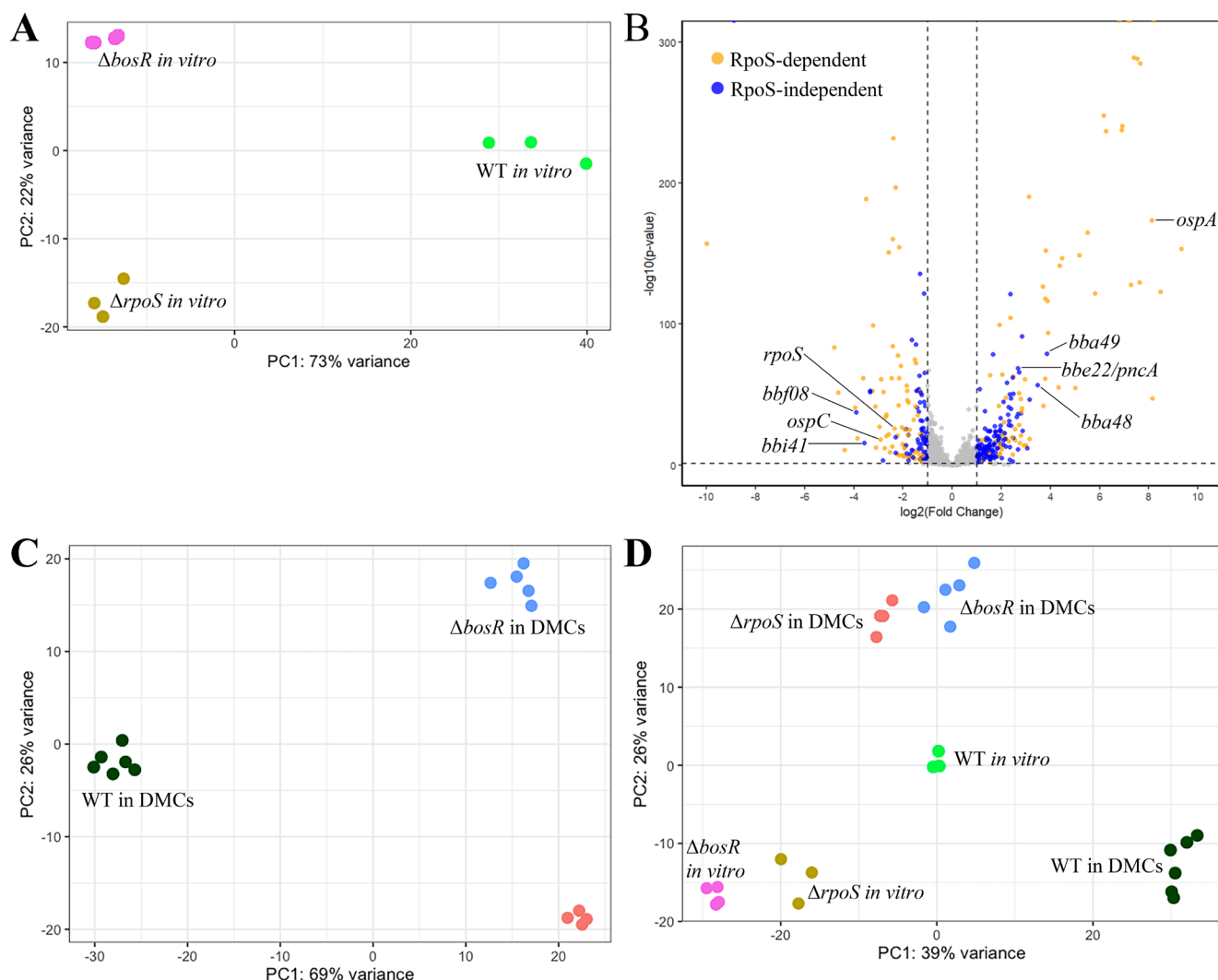
## 2 | Results

### 2.1 | The BosR Regulon In Vitro

Since the discovery that BosR is required for RpoN-dependent expression of *rpoS*, most research has centered on its pivotal role within the RpoN/RpoS regulatory pathway (Grassmann et al. 2023; Hyde et al. 2009, 2010; Katona 2015; Mason et al. 2019; Ouyang et al. 2011, 2009, 2015; Raghunandanan et al. 2024; Seshu et al. 2004; Shi et al. 2014; Sze and Li 2023; Wang et al. 2013). However, the original studies describing BosR as a regulator of *B. burgdorferi*'s oxidative stress response involved genes now known to reside outside of the RpoS regulon (Boylan et al. 2003; Seshu et al. 2004; Hyde et al. 2006, 2009; Ouyang et al. 2009; Esteve-Gassent et al. 2009; Hyde et al. 2010; Wang et al. 2013). To delineate the presumptive RpoS-dependent and -independent components of the BosR regulon, we generated a  $\Delta$ *bosR* strain in the WT B31 5A4 background by replacing the native *bosR* (*bb0647*) locus with a *PflgB-kan* kanamycin resistance cassette. We selected a clone retaining all endogenous plasmids and confirmed the mutation by sequencing and immunoblot analysis. The  $\Delta$ *bosR* mutant displayed a modest but not statistically significant ( $p = 0.11$ ) growth defect under microaerobic in vitro conditions at 37°C after temperature shift (Figure S1A), consistent with the phenotype previously reported by Hyde et al. (2009). As expected, BosR, RpoS, and the prototypical RpoS-dependent gene products OspC and DbpA (Eggers et al. 2004; Hubner et al. 2001) were absent in immunoblots of in vitro-cultivated  $\Delta$ *bosR* (Figure S1B). We next defined the in vitro BosR regulon by RNA-seq, comparing the transcriptomes of WT and  $\Delta$ *bosR* following a temperature shift in BSK-II medium. A summary of mapped reads and SRA accession numbers for all RNA-seq samples analyzed in this study is provided in Table S1 under BioProject PRJNA1307481. To identify RpoS-regulated genes within the BosR regulon, we re-analyzed our previously published (Caimano et al. 2019) RNA-seq dataset comparing isogenic WT and  $\Delta$ *rpoS* strains cultivated under the same in vitro conditions used herein. We derived the RpoS-independent subset of the in vitro BosR regulon by excluding *rpoS* itself and all RpoS-regulated genes.

We first performed Principal Component Analysis (PCA) of the RNA-seq data to gain an overview of global transcriptional differences between the two strains (Figure 2A). The PCA clearly separated the in vitro-cultivated WT,  $\Delta$ *bosR*, and  $\Delta$ *rpoS* strains, indicating that BosR modulates transcriptional responses in an RpoS-dependent as well as RpoS-independent manner. A total of 459 genes were significantly dysregulated ( $\geq 2$ -fold,  $q < 0.05$ ) in  $\Delta$ *bosR* compared to WT in vitro (Table S2), including *rpoS* and 279 genes belonging to the RpoS regulon, with prototypical RpoS-dependent genes (Eggers et al. 2004; Grassmann et al. 2023; Caimano et al. 2019) strongly downregulated at the transcriptional level. After excluding all RpoS-regulated genes, we identified 176 genes (94 activated and 82 repressed) regulated by BosR independently of RpoS. Only eight of these RpoS-independent genes overlapped with genes previously identified in a microarray-based assessment of the in vitro BosR regulon by Ouyang et al. (2009). Remarkably, expression of oxidative stress genes previously linked to BosR—*napA/dps*





**FIGURE 2** | BosR regulates *Borrelia burgdorferi* gene expression through RpoS-dependent and -independent mechanisms. (A) PCA of RNA-seq data from WT,  $\Delta$ *bosR*, and  $\Delta$ *rpoS* strains cultivated in vitro shows clear separation of the BosR and RpoS regulons. (B) Volcano plot comparing WT and  $\Delta$ *bosR* transcriptomes in DMCs highlights BosR-regulated genes in host-adapted *B. burgdorferi*. RpoS-dependent (blue) and RpoS-independent (orange) BosR-regulated genes are distinguished, and representative genes of interest are labeled. (C) PCA of DMC-cultivated WT,  $\Delta$ *bosR*, and  $\Delta$ *rpoS* transcriptomes shows distinct clustering of each strain. (D) PCA of combined datasets show that the in vitro and DMC-cultivated  $\Delta$ *bosR* and  $\Delta$ *rpoS* form well separated clusters. The transcriptomes of WT spirochetes cultivated in vitro and in DMCs also separated clearly.

(Hyde et al. 2010; Boylan et al. 2003; Seshu et al. 2004; Hyde et al. 2009), *sodA* (Hyde et al. 2006, 2010; Seshu et al. 2004; Esteve-Gassent et al. 2009) and *cdr* (Hyde et al. 2009)—was not affected by loss of BosR under the in vitro growth conditions employed herein (Table 1). Instead, BosR modulated expression of a small set of genes that promote DNA repair and redox homeostasis (Table 1). BosR activated *bb0797/mutS*, a mismatch repair protein critical for oxidative DNA damage repair and reactive oxygen species (ROS) and reactive nitrogen species (RNS) resistance (Bourret et al. 2016; Troxell et al. 2014). On the other hand, BosR repressed *bb0344/uvrD* and activated *bb0839*. *uvrD* encodes a DNA helicase involved in nucleotide excision and mismatch repair that counters ROS and RNS (Bourret et al. 2016; Hardy and Chaconas 2013), while *bb0839* encodes an RNase HI-like protein that contributes to nitric oxide resistance (Ramsey et al. 2017). BosR also repressed *bb0164*, encoding a membrane protein central to manganese homeostasis and ROS defense (Ramsey et al. 2017)

and *bb0318/mglA*, encoding the ATPase component of a riboflavin transporter essential for redox cofactor biosynthesis and oxidative stress tolerance (Showman et al. 2016).

## 2.2 | BosR Regulates Genes Independently of RpoS in Mammalian Host-Adapted Spirochetes

To date, most studies on gene regulation by BosR have been conducted in vitro (Boylan et al. 2003; Hyde et al. 2006, 2009, 2010; Katona 2015; Mason et al. 2019; Ouyang et al. 2011, 2009, 2015; Raghunandan et al. 2024; Seshu et al. 2004; Shi et al. 2014; Sze and Li 2023; Wang et al. 2013, 2017), conditions that poorly reflect the transcriptomic state of *B. burgdorferi* within its natural tick or mammalian hosts (Akins et al. 1998; Caimano 2018; Caimano et al. 2005, 2019, 2007; Grassmann et al. 2023; Grove et al. 2017; Iyer et al. 2015). To better define the BosR regulon under more physiologically relevant conditions, we

**TABLE 1** | Expression of genes associated with oxidative and/or nitrosative stress responses in WT versus *ΔbosR* strains under in vitro and mammalian host-adapted conditions.

| Locus tag <sup>a</sup> | Gene <sup>a</sup> | Product <sup>a</sup>                                                              | Previously linked to BosR? <sup>b</sup> | Fold change WT vs. <i>ΔbosR</i> in vitro <sup>c</sup> | RpoS-dependent in vitro? <sup>d</sup> | Fold change WT vs. <i>ΔbosR</i> in DMCs <sup>c</sup> | RpoS-dependent in DMCs? <sup>d</sup> | Linked to ROS or RNS response? <sup>e</sup> | References                                                                  |
|------------------------|-------------------|-----------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------|---------------------------------------|------------------------------------------------------|--------------------------------------|---------------------------------------------|-----------------------------------------------------------------------------|
| BB0153                 | <i>sodA</i>       | Superoxide dismutase                                                              | Yes                                     | −1.31                                                 | No                                    | 1.17                                                 | No                                   | ROS                                         | (Hyde et al. 2006, 2010; Seshu et al. 2004; Esteve-Gassent et al. 2009)     |
| BB0690                 | <i>napA</i>       | Neutrophil activating protein A (Ferritin/DPS)                                    | Yes                                     | 1.16                                                  | No                                    | −1.37                                                | No                                   | ROS                                         | (Hyde et al. 2010; Boylan et al. 2003; Seshu et al. 2004; Hyde et al. 2009) |
| BB0728                 | <i>cdr</i>        | CoA-disulfide reductase                                                           | Yes                                     | 1.32                                                  | Yes <sup>f</sup>                      | 1.50                                                 | No                                   | ROS                                         | (Hyde et al. 2009)                                                          |
| BB0344                 | <i>uvrD</i>       | DNA helicase                                                                      | Yes                                     | −2.15                                                 | Yes                                   | 2.28                                                 | No                                   | RNS                                         | (Ramsey et al. 2017; Hyde et al. 2006; Hardy and Chaconas 2013)             |
| BB0839                 | —                 | Ribonuclease H1 N-terminal domain-containing protein                              | Yes                                     | 2.03                                                  | No                                    | −1.52                                                | No                                   | RNS                                         | (Ramsey et al. 2017; Hyde et al. 2006)                                      |
| BB0164                 | —                 | K <sup>+</sup> -dependent Na <sup>+</sup> /Ca <sup>+</sup> exchanger-like protein | No                                      | −2.20                                                 | No                                    | 1.24                                                 | No                                   | ROS                                         | (Ramsey et al. 2017)                                                        |
| BB0318                 | <i>mgIA</i>       | Methylgalactoside ABC transporter ATP-binding protein                             | No                                      | −2.37                                                 | Yes                                   | 1.23                                                 | No                                   | ROS                                         | (Showman et al. 2016)                                                       |
| BB0797                 | <i>mutS</i>       | DNA mismatch repair protein                                                       | No                                      | 2.71                                                  | Yes                                   | 2.48                                                 | No                                   | ROS                                         | (Bourret et al. 2016; Troxell et al. 2014)                                  |
| BB0017                 | —                 | Integral membrane protein                                                         | No                                      | −1.57                                                 | No                                    | 1.76                                                 | No                                   | ROS                                         | (Ramsey et al. 2017)                                                        |
| BB0202                 | —                 | Hemolysin                                                                         | No                                      | −1.15                                                 | No                                    | −1.52                                                | No                                   | ROS                                         | (Ramsey et al. 2017)                                                        |
| BB0457                 | <i>uvrC</i>       | Excinuclease ABC subunit C                                                        | No                                      | 1.00                                                  | No                                    | 1.10                                                 | No                                   | RNS                                         | (Ramsey et al. 2017; Bourret et al. 2011; Hardy and Chaconas 2013)          |
| BB0836                 | <i>uvrB</i>       | Excinuclease ABC subunit B                                                        | No                                      | −1.01                                                 | No                                    | 1.91                                                 | No                                   | ROS/RNS                                     | (Bourret et al. 2016, 2011; Troxell et al. 2014; Hardy and Chaconas 2013)   |
| BB0837                 | <i>uvrA</i>       | Excinuclease ABC subunit A                                                        | No                                      | −1.25                                                 | No                                    | 1.80                                                 | No                                   | ROS                                         | (Sambir et al. 2011; Hardy and Chaconas 2013)                               |

<sup>a</sup>Locus tags, gene names, and product descriptions are derived from the *B. burgdorferi* strain B31 RefSeq genome annotation and/or UniProt/InterPro analyses.

<sup>b</sup>Yes = Gene expression previously linked to BosR in published studies.

<sup>c</sup>Fold changes in gene expression were determined by comparing WT to *ΔbosR* strains in vitro (Table S2) or in DMCs (Table S3). Only genes showing ≥ 2-fold differences with  $q < 0.05$  were considered significantly regulated. Green highlights indicate genes activated by BosR; yellow highlights indicate genes repressed by BosR.

<sup>d</sup>Yes = Gene is part of the RpoS regulon under in vitro or DMC conditions, as previously defined by RNA-seq analysis from our group (Caimano et al. 2019; Grassmann et al. 2023).

<sup>e</sup>Genes previously linked to *B. burgdorferi*'s response to reactive oxygen species (ROS) or reactive nitrogen species (RNS), as documented in the references listed in the final column.

<sup>f</sup>*cdr* is an RpoS-dependent gene in vitro and dually transcribed by RpoS and RpoD in DMCs (Caimano et al. 2019; Grassmann et al. 2023).

**TABLE 2** | Selected RpoS-regulated genes also regulated by BosR in mammalian host-adapted (DMC-cultivated) *Borrelia burgdorferi*.

| Locus tag <sup>b</sup>                        | Gene <sup>b</sup> | Product <sup>b</sup>                                                          | Fold change in WT vs. $\Delta$ <i>bosR</i> <sup>c</sup> | Fold change in WT vs. <i>bosR-R39A</i> $\Delta$ <i>rpoS/irpoS</i> <sup>d</sup> |
|-----------------------------------------------|-------------------|-------------------------------------------------------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------|
| Genes activated by RpoS and BosR <sup>a</sup> |                   |                                                                               |                                                         |                                                                                |
| BBF0041                                       | <i>vlsE1</i>      | Outer surface protein VlsE1                                                   | 1006.21                                                 | 1.28                                                                           |
| BBA25                                         | <i>dbpB</i>       | Decorin binding protein B                                                     | 27.58                                                   | 2.97                                                                           |
| BBA34                                         | <i>oppA5</i>      | Extracellular solute-binding protein OppA5                                    | 9.28                                                    | 6.41                                                                           |
| BBK32                                         | —                 | Fibronectin-binding protein                                                   | 8.74                                                    | 3.03                                                                           |
| BBB19                                         | <i>ospC</i>       | Outer surface protein C                                                       | 7.50                                                    | 1.75                                                                           |
| BBA24                                         | <i>dbpA</i>       | Decorin binding protein A                                                     | 6.45                                                    | 2.95                                                                           |
| BB0681                                        | <i>mcp5</i>       | Methyl-accepting chemotaxis protein Mcp5                                      | 5.35                                                    | 2.55                                                                           |
| BB0680                                        | <i>mcp4</i>       | Methyl-accepting chemotaxis protein Mcp4                                      | 4.44                                                    | 2.12                                                                           |
| BB0669                                        | <i>cheA2</i>      | Chemotaxis protein CheA2                                                      | 2.94                                                    | 2.51                                                                           |
| Genes repressed by RpoS and BosR <sup>a</sup> |                   |                                                                               |                                                         |                                                                                |
| BBJ09                                         | <i>ospD</i>       | Outer surface protein D                                                       | −650.90                                                 | −581.81                                                                        |
| BBA74                                         | <i>bba74</i>      | Osm28                                                                         | −296.74                                                 | −170.51                                                                        |
| BBA15                                         | <i>ospA</i>       | Outer surface protein A                                                       | −281.17                                                 | −152.92                                                                        |
| BBA68                                         | <i>BBCRASP-1</i>  | Complement regulator-acquiring surface protein 1                              | −203.17                                                 | −46.49                                                                         |
| BBA16                                         | <i>ospB</i>       | Outer surface protein B                                                       | −187.10                                                 | −100.20                                                                        |
| BBA62                                         | <i>bba62</i>      | 6.6 kDa lipoprotein                                                           | −156.60                                                 | −80.01                                                                         |
| BBI39                                         | —                 | PFam_54/60                                                                    | −144.65                                                 | −62.16                                                                         |
| BBI38                                         | —                 | PFam_54/60                                                                    | −20.16                                                  | −44.43                                                                         |
| BBD18                                         | —                 | BBD18                                                                         | −8.05                                                   | −13.10                                                                         |
| BB0240                                        | <i>glpF</i>       | Glycerol uptake facilitator                                                   | −7.01                                                   | −13.41                                                                         |
| BBB29                                         | <i>malX2</i>      | PTS system transporter subunit IIBC                                           | −4.96                                                   | −1.93                                                                          |
| BB0241                                        | <i>glpK</i>       | Glycerol kinase                                                               | −4.58                                                   | −10.00                                                                         |
| BB0330                                        | <i>oppA3</i>      | Oligopeptide ABC transporter periplasmic oligopeptide-binding protein (OppA3) | −3.72                                                   | −5.93                                                                          |
| BB0243                                        | <i>glpD</i>       | Glycerol-3-phosphate dehydrogenase                                            | −2.93                                                   | −5.42                                                                          |

<sup>a</sup>Genes were selected based on their predicted or experimentally validated functions and prior identification as RpoS-regulated in multiple studies. A complete list of RpoS-regulated genes within the mammalian-phase BosR regulon is provided in Table S3.

<sup>b</sup>Locus tags, gene names, and product descriptions are derived from the *B. burgdorferi* strain B31 RefSeq genome annotation and/or UniProt/InterPro analyses.

<sup>c</sup>Fold changes in gene expression were determined by comparing WT to  $\Delta$ *bosR* strains in DMCs (Table S3). Only genes showing  $\geq 2$ -fold differences with  $q < 0.05$  were considered significantly regulated. All genes shown meet the  $q < 0.05$  threshold.

<sup>d</sup>Fold changes in gene expression were determined by comparing WT to *bosR-R39A* $\Delta$ *rpoS/irpoS* + IPTG strains in DMCs (Table S3). Only genes showing  $\geq 2$ -fold differences with  $q < 0.05$  were considered significantly regulated. All genes shown meet the threshold, except those shaded in pale peach color.

performed RNA-seq on WT and  $\Delta$ *bosR* strains cultivated within DMCs implanted intraperitoneally in rats (Akins et al. 1998; Caimano 2018). A volcano plot comparing WT and  $\Delta$ *bosR* transcriptomes in DMCs (Figure 2B) highlights the broad scope of BosR-dependent regulation, typical of global regulators. In

DMCs, BosR regulated 364 genes ( $\geq 2$ -fold,  $q < 0.05$ ; Table 2, Table 3, Table S3), including *rpoS* and 146 genes (73 activated and 73 repressed) from the previously defined mammalian-phase RpoS regulon (Grassmann et al. 2023) (Table 2). The activated RpoS-dependent genes within the BosR regulon represent

**TABLE 3** | Selected genes regulated by BosR independently of RpoS in mammalian host-adapted (DMC-cultivated) *Borrelia burgdorferi*.

| Locus tag <sup>b</sup>                                | Gene <sup>b</sup> | Product <sup>b</sup>                                                  | Functional category <sup>c</sup>                | Fold change in WT vs. $\Delta$ <i>bosR</i> <sup>d</sup> | Fold change in WT vs. <i>bosR-R39A</i> <sup>e</sup> |
|-------------------------------------------------------|-------------------|-----------------------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------|
| RpoS-independent genes activated by BosR <sup>a</sup> |                   |                                                                       |                                                 |                                                         |                                                     |
| BBF08                                                 | —                 | Uncharacterized protein                                               | Uncharacterized protein                         | 14.94                                                   | 7.42                                                |
| BBI41                                                 | —                 | Transposase putative helix-turn-helix domain-containing protein       | Mobilome: prophages, transposons                | 11.89                                                   | 7.73                                                |
| BBQ89                                                 | —                 | Lipoprotein                                                           | Lipoprotein                                     | 10.11                                                   | 6.75                                                |
| BBD001                                                | —                 | Lipoprotein                                                           | Lipoprotein                                     | 9.98                                                    | 6.86                                                |
| BBJ0058                                               | —                 | Uncharacterized protein                                               | Uncharacterized protein                         | 7.01                                                    | 13.66                                               |
| BBI34                                                 | —                 | PFam_54/60                                                            | Lipoprotein                                     | 4.92                                                    | 5.19                                                |
| BBI06                                                 | —                 | MTA/SAH nucleosidase                                                  | Nucleotide transport and metabolism             | 4.91                                                    | 4.80                                                |
| BBI22                                                 | <i>pf49</i>       | PF-49 protein                                                         | Chromosome partitioning                         | 3.58                                                    | 4.40                                                |
| BB0568                                                | <i>cheB2</i>      | Chemotaxis response regulator CheB                                    | Signal transduction mechanisms                  | 2.56                                                    | 2.24                                                |
| BB0414                                                | <i>cheR2</i>      | Chemotaxis protein methyltransferase CheR-2                           | Signal transduction mechanisms                  | 2.54                                                    | 1.67                                                |
| BB0797                                                | <i>mutS</i>       | DNA mismatch repair protein MutS                                      | Replication, recombination and repair           | 2.48                                                    | 2.24                                                |
| BB0343                                                | <i>gatC</i>       | Glutamyl-tRNA (Gln) amidotransferase subunit C                        | Translation, ribosomal structure and biogenesis | 2.45                                                    | 2.85                                                |
| BB0729                                                | <i>gltP</i>       | Dicarboxylate/Amino acid: cation symporter                            | Amino acid transport and metabolism             | 2.36                                                    | 2.36                                                |
| BB0420                                                | <i>hk1</i>        | Sensory transduction histidine kinase                                 | Signal transduction mechanisms                  | 2.33                                                    | 2.07                                                |
| BB0344                                                | <i>uvrD</i>       | DNA helicase UvrD                                                     | Replication, recombination and repair           | 2.28                                                    | 1.55                                                |
| BB0312                                                | <i>cheW1</i>      | Purine-binding chemotaxis protein CheW                                | Signal transduction mechanisms                  | 2.19                                                    | 1.72                                                |
| BB0329                                                | <i>oppA2</i>      | ABC-type oligopeptide transport system, periplasmic component (OppA2) | Amino acid transport and metabolism             | 2.10                                                    | 1.41                                                |
| BBI21                                                 | <i>pf32</i>       | PF-32 protein                                                         | Chromosome partitioning                         | 2.09                                                    | 3.69                                                |
| BB0693                                                | <i>badR</i>       | Xylose operon regulatory protein, BadR                                | Carbohydrate transport and metabolism           | 2.00                                                    | 1.65                                                |
| RpoS-independent genes repressed by BosR <sup>a</sup> |                   |                                                                       |                                                 |                                                         |                                                     |
| BBA48                                                 | —                 | DUF1473 family protein                                                | Uncharacterized protein                         | −14.52                                                  | −10.14                                              |
| BBA49                                                 | —                 | DUF1322 family protein                                                | Uncharacterized protein                         | −11.28                                                  | −9.31                                               |
| BBA51                                                 | —                 | DUF792 family protein                                                 | Uncharacterized protein                         | −8.90                                                   | −6.50                                               |
| BBQ09                                                 | —                 | DUF226 domain-containing protein                                      | Uncharacterized protein                         | −8.36                                                   | −8.42                                               |
| BBP38                                                 | <i>erpA</i>       | Protein ErpA                                                          | Lipoprotein                                     | −7.34                                                   | −4.99                                               |

(Continues)

TABLE 3 | (Continued)

| Locus tag <sup>b</sup> | Gene <sup>b</sup> | Product <sup>b</sup>                                                  | Functional category <sup>c</sup>       | Fold change in WT vs. $\Delta$ <i>bosR</i> <sup>d</sup> | Fold change in WT vs. <i>bosR-R39A</i> <sup>e</sup> |
|------------------------|-------------------|-----------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------|-----------------------------------------------------|
| BBB23                  | —                 | Guanine/Xanthine permease                                             | Nucleotide transport and metabolism    | −7.03                                                   | −7.34                                               |
| BBE22                  | <i>pncA</i>       | Pyrazinamidase/Nicotinamidase PncA                                    | Coenzyme transport and metabolism      | −6.45                                                   | −6.57                                               |
| BBE16                  | <i>bptA</i>       | Virulence-associated protein BptA                                     | Lipoprotein                            | −5.66                                                   | −5.01                                               |
| BBB16                  | <i>oppA4</i>      | ABC-type oligopeptide transport system, periplasmic component (OppA4) | Amino acid transport and metabolism    | −5.52                                                   | −5.86                                               |
| BBA30                  | —                 | Helix-turn-helix domain-containing protein                            | Uncharacterized protein                | −4.85                                                   | −3.57                                               |
| BBP39                  | <i>erpB</i>       | Protein ErpB                                                          | Lipoprotein                            | −4.60                                                   | −2.46                                               |
| BBL28                  | <i>mlpH</i>       | MlpH lipoprotein                                                      | Lipoprotein                            | −4.26                                                   | −2.26                                               |
| BBN28                  | <i>mlpI</i>       | MlpI lipoprotein                                                      | Lipoprotein                            | −3.67                                                   | −1.58                                               |
| BBN38                  | <i>erpP</i>       | Protein ErpP                                                          | Lipoprotein                            | −3.64                                                   | −2.16                                               |
| BBH06                  | <i>cpsZ</i>       | Complement regulator-acquiring surface protein CpsZ                   | Lipoprotein                            | −3.33                                                   | −3.09                                               |
| BBN39                  | <i>erpQ</i>       | Protein ErpQ                                                          | Lipoprotein                            | −2.83                                                   | −2.99                                               |
| BB0785                 | <i>spoVG</i>      | Septation protein SpoVG                                               | Chromosome partitioning                | −2.77                                                   | −1.60                                               |
| BBB22                  | —                 | Guanine/Xanthine permease                                             | Nucleotide transport and metabolism    | −2.75                                                   | −2.89                                               |
| BBE31                  | <i>bbe31</i>      | PFam_54/60                                                            | Lipoprotein                            | −2.67                                                   | −3.44                                               |
| BB0215                 | <i>pstS</i>       | Phosphate ABC transporter substrate-binding protein                   | Inorganic ion transport and metabolism | −2.06                                                   | −1.74                                               |

<sup>a</sup>Genes were selected based on their predicted or experimentally validated functions and the magnitude of their regulation (fold-change). A complete list of RpoS-independent genes within the mammalian-phase BosR regulon is provided in Table S3.

<sup>b</sup>Locus tags, gene names, and product descriptions are derived from the *B. burgdorferi* strain B31 RefSeq genome annotation and/or UniProt/InterPro analyses.

<sup>c</sup>Functional categories were assigned based on COG terms provided by InterPro and refined by manual curation.

<sup>d</sup>Fold changes in gene expression were determined by comparing WT to  $\Delta$ *bosR* strains in DMCs (Table S3). Only genes showing  $\geq 2$ -fold differences with  $q < 0.05$  were considered significantly regulated. All genes shown meet the  $q < 0.05$  threshold.

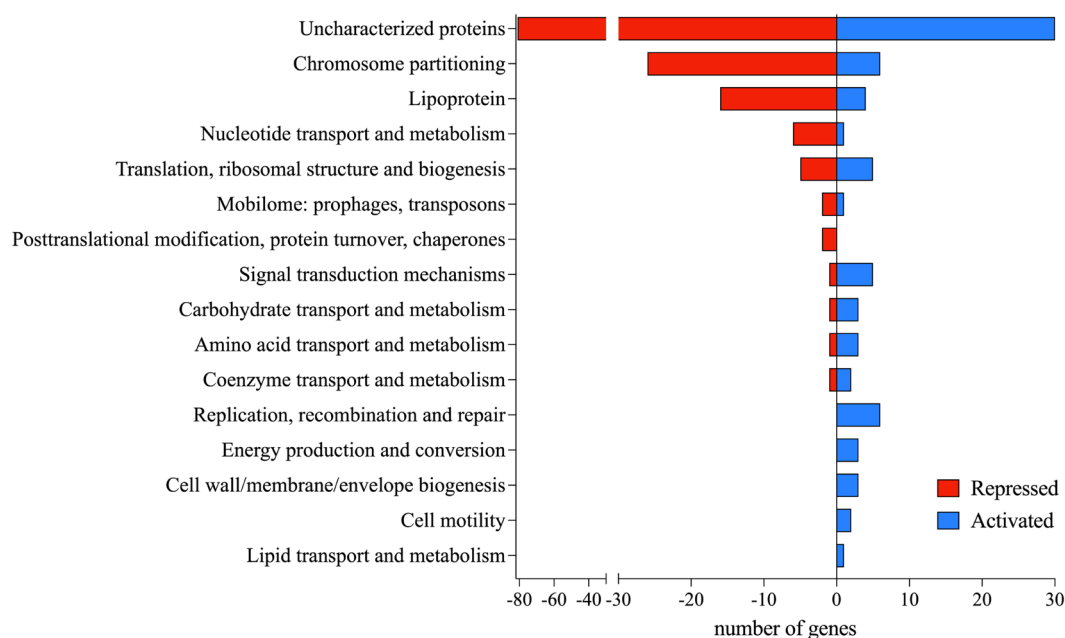
<sup>e</sup>Fold changes in gene expression were determined by comparing WT to *bosR-R39A* strains in DMCs (Table S3). Only genes showing  $\geq 2$ -fold differences with  $q < 0.05$  were considered significantly regulated. All genes shown meet the threshold, except those shaded in pale peach color.

~80% of the RpoS regulon in mammals (Grassmann et al. 2023) and include prototypical genes such as *ospC*, *dbpBA*, *bbk32*, and *mcp4/5* (Grassmann et al. 2023; Caimano et al. 2019, 2007). In addition to activating mammalian-phase genes, in host-adapted *B. burgdorferi* RpoS also functions as a transcriptional repressor of numerous tick-phase genes, such as *ospAB*, *glpDFK*, *bba74*, and *bba62* (Akins et al. 1998; Caimano et al. 2004, 2007; Grassmann et al. 2023; Grove et al. 2017). In ticks, these genes are expressed at high levels by RNAP-RpoD (Grassmann et al. 2023; Iyer et al. 2015; Sapiro et al. 2023) but are silenced by RNAP-RpoS in collaboration with BosR (Grassmann et al. 2023) upon mammalian host adaptation. Loss of *rpoS* expression in the  $\Delta$ *bosR* mutant in DMCs resulted in de-repression of ~90% (73 genes) of the RpoS-repressed genes, including all prototypical repressed tick-phase genes.

The PCA of DMC-cultivated WT,  $\Delta$ *bosR*, and  $\Delta$ *rpoS* spirochetes (Figure 2C) revealed that  $\Delta$ *bosR* clustered distinctly from both

WT and  $\Delta$ *rpoS*, indicating, as with in vitro-cultivated spirochetes, that a substantial portion of the in vivo BosR regulon resides outside of the cohort of genes controlled by RpoS. Accordingly, we identified 217 genes (75 activated, 142 repressed) regulated by BosR independently of RpoS in mammalian host-adapted *B. burgdorferi* (Table 3, Table S3). Interestingly, most of these RpoS-independent genes showed modest changes in expression, with 78 of 142 (~55%) repressed and 58 of 75 (~77%) activated genes differing by only 2–3-fold in  $\Delta$ *bosR* in DMCs, suggesting a fine-tuning role for BosR. As with the in vitro transcriptome, expression of previously identified putative BosR-regulated oxidative stress response genes (*napA/dps*, *sodA*, and *cdr*) was not affected by loss of BosR in  $\Delta$ *bosR* under in vivo conditions (Table 1). Only two genes previously associated with oxidative and/or nitrosative stress responses—*uvrD* and *mutS*—were regulated (both activated) by BosR in DMCs (Table 1), suggesting that BosR contributes minimally to the defense against oxidative and nitrosative stress within the mammalian host.





**FIGURE 3** | Functional categorization of RpoS-independent genes in the mammalian-phase BosR regulon. BosR-regulated, RpoS-independent genes in DMC-cultivated spirochetes were assigned to functional categories based on InterProScan annotation of their encoded protein domains and predicted features.

Although  $\Delta\text{bosR}$  and  $\Delta\text{rpoS}$  samples separated in individual PCAs of the in vitro and DMC datasets (Figure 2A,C), a combined PCA of both datasets (Figure 2D) revealed that DMC-cultivated  $\Delta\text{bosR}$  and  $\Delta\text{rpoS}$  clustered more closely with each other than with their in vitro counterparts. This clustering pattern highlights substantial differences in the BosR regulon under in vitro and in vivo conditions. Consistent with the condition-specific regulatory activity of BosR, only five RpoS-independent genes (*bdd04*, *bbf08*, *bbi18*, *bbi085*, and *bbq0091*), all encoding hypothetical proteins, were dysregulated in both in vitro- and DMC-cultivated  $\Delta\text{bosR}$ . Notably, WT spirochetes cultivated in vitro and in DMCs also separated widely, underscoring the genome-wide transcriptional reprogramming that accompanies mammalian host adaptation (Grassmann et al. 2023; Iyer et al. 2015). Presumably, the broad separation between the in vitro and DMC BosR and RpoS clusters also reflects BosR- and RpoS-independent genes differentially expressed across these environments.

*Genes activated by BosR independently of RpoS in mammalian host-adapted spirochetes* (Figure 3, Table 3, Table S3). We used InterProScan to annotate the proteins encoded by the RpoS-independent genes in the BosR regulon and inferred functions based on conserved domains (Figure 3). Nearly half (30 of 75) of the RpoS-independent genes activated by BosR in mammalian host-adapted organisms encode uncharacterized proteins, defined as those with little or no experimental evidence supporting their function, structure, or biological roles. BosR also activated four genes for lipoproteins of unknown function: *bbi34*, which encodes a member of the Pfam54\_60 family (Brangulis et al. 2020), the identical genes *bdd001* and *bbq89*, which encode a short lipoprotein (~6 kDa), and *bdd10*. Notably, BosR activated *bb0729/gltP*, which encodes a dicarboxylate amino acid/ $\text{Na}^+$ /L-cystine symporter (Eggers et al. 2011) co-transcribed with *bb0728/cdr* (Eggers et al. 2011). Interestingly,

*bb0728/cdr* was not regulated by BosR in vivo (Table 1), pointing to differential regulation within the *gltP-cdr* operon. BosR also activated *badR*, which encodes a xylose operon regulator (Zhang et al. 2024; Samuels et al. 2021; Ouyang and Zhou 2015; Miller et al. 2013). In vitro, BadR binds to the promoters of *bosR* and *rpoS*, repressing their expression (Ouyang and Zhou 2015; Miller et al. 2013; George and Ouyang 2025). Recent work defining the BadR regulon under in vitro conditions (George and Ouyang 2025) identified 75 RpoS-dependent and 64 RpoS-independent genes that are also regulated by BosR in mammalian host-adapted spirochetes (Table S3). While the comparison involves datasets generated under different environmental conditions, the extent of overlap suggests a potential functional relationship between the BosR and BadR regulons. Interestingly, BosR also activated *bb0420/hk1*, encoding the sensory transduction histidine kinase Hk1. Although *hk1* is expressed in mammals, activation of the Hk1/Rrp1 two-component system and subsequent c-di-GMP production is restricted to the tick phase of the enzootic cycle (Kostick et al. 2011; He et al. 2011; Groshong, Grassmann, et al. 2021). Finally, BosR activated the chemotaxis-related genes *bb0312/cheW1*, *bb0568/cheB2*, and *bb0414/cheR* (Charon et al. 2012; Zhang et al. 2012), as well as *oppA2*, which encodes one of the spirochete's five periplasmic oligopeptide-binding proteins (OBPs) (Lin et al. 2001; Wang et al. 2004; Groshong et al. 2017).

*Genes repressed by BosR independently of RpoS in mammalian host-adapted spirochetes* (Figure 3, Table 3, Table S3). Most RpoS-independent genes repressed by BosR in mammalian host-adapted *B. burgdorferi* (81 of 142; ~57%) encode uncharacterized proteins. Curiously, several BosR-repressed genes with known functions are involved in mammalian infection. These include *cspZ/bbh06*, which encodes a factor H-binding lipoprotein that promotes complement resistance (Hartmann et al. 2006; Lin et al. 2020), and the purine permeases encoded

by *bbb22* and *bbb23*, both essential for mammalian infection (Jain et al. 2012). BosR also repressed *bbe22/pncA*, encoding a nicotinamidase required for infectivity (Purser et al. 2003). Mapped DESeq2 counts for these mammalian-phase genes in WT spirochetes revealed that even with BosR-mediated repression they are expressed at relatively high levels in mammalian host-adapted *B. burgdorferi* (Table S4). Among the most strongly repressed genes were *bba48*, *bba49* and *bba51*, all encoded in the linear plasmid lp54, and *bbe16/bptA*, encoding BptA (borreliar persistence in ticks protein A), a persistence factor required for survival in ticks (Gilmore et al. 2014; Revel et al. 2005; Jewett et al. 2011; Purser et al. 2003). In addition to *cspZ/bbh06*, BosR repressed 15 other putative lipoproteins in DMCs, including *pstS/bb0215*, *bbe31*, *erpA/bbp38*, *erpB/bbp39*, *erpP/bbn38*, *erpQ/bbn39*, *mlpH/bbl28*, and *mlpI/bbn28*. PstS is a periplasmic substrate-binding lipoprotein involved in inorganic phosphate transport (Brautigam et al. 2014), and BBE31 is a lipoprotein essential for transmission (Zhang et al. 2011) whose expression is RpoS-dependent in nymphs (Grassmann et al. 2023). The cp32-encoded genes annotated as 'erps' belong to three evolutionarily distinct outer surface lipoprotein families—OspE, OspF, and Elps (Caimano et al. 2000; Akins et al. 1999; Lin et al. 2015). *erpA/bbp38* and *erpP/bbn38* are OspEs, while *erpB/bbp39* and *erpQ/bbn39* are Elps; OspE and Elp lipoproteins play key roles in adhesion (Stevenson and Brissette 2023; Brissette et al. 2008) and complement evasion (Hellwage et al. 2001; Kraiczy et al. 2001; Bhattacharjee et al. 2013; Garrigues et al. 2022; Thomas et al. 2024; Hill et al. 2025). Also encoded in cp32s, the Mlp (multicopy lipoproteins) surface lipoproteins have undefined function, but their expression patterns and antigenicity suggest roles during vertebrate infection, potentially involving adhesion to host cells and tissues (Yang et al. 1999, 2003; Stevenson and Brissette 2023). BosR also repressed 26 genes involved in plasmid/chromosome partitioning, including multiple *pf32* genes on various plasmids, indicating a hitherto unknown role in cell cycle regulation. Additional BosR-repressed genes included *spoVG* (Jutras et al. 2013; Savage et al. 2018; Saylor et al. 2023), encoding a DNA-binding protein; *bba30* (Bhatia et al. 2021), encoding a putative DNA-binding protein with a predicted helix-turn-helix domain but unconfirmed binding activity, as well as *bbb16/oppA4*, an OBP whose expression is RpoS-independent in mammals (Grassmann et al. 2023). Notably, Medrano et al. previously showed that BosR binds the *oppA4* promoter (Medrano et al. 2007), while Zhou et al. (2018) reported that OppA4 negatively regulates *bosR* expression at the transcriptional level (Zhou et al. 2018). Taken together, these findings support the existence of a negative feedback loop between BosR and OppA4.

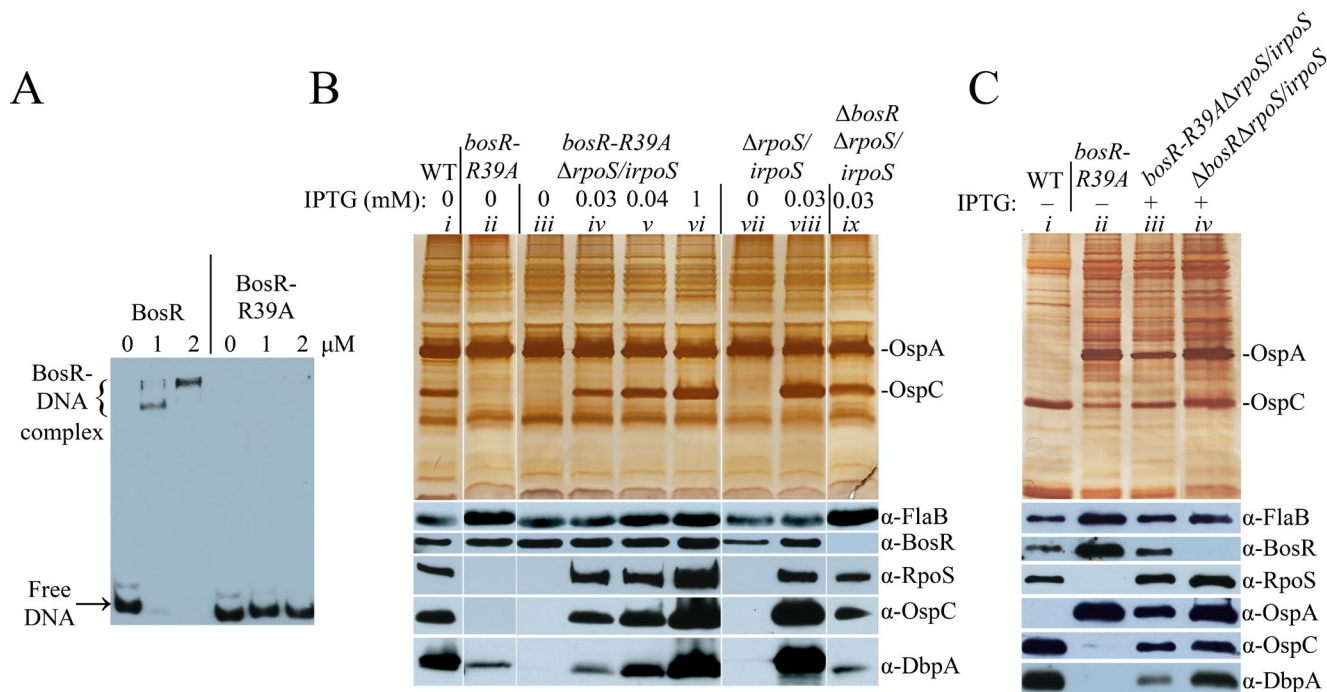
### 2.3 | DNA-Binding by BosR is Essential for Regulation of RpoS-Dependent and -Independent Genes

BosR contains a FUR-family DNA-binding domain that includes a highly conserved arginine at position 39 (R39) (Katona 2015; Seshu et al. 2004; Grassmann et al. 2023; Sarvan, Butcher, et al. 2018; Sevilla et al. 2021; Yang et al. 2022; Liu et al. 2021) (Figure 1). Prior studies showed that a naturally occurring R39K substitution impairs DNA binding by BosR, reducing the expression of *rpoS* (Hyde et al. 2006; Katona 2015). Because lysine preserves a positive charge, we engineered an R39A mutation

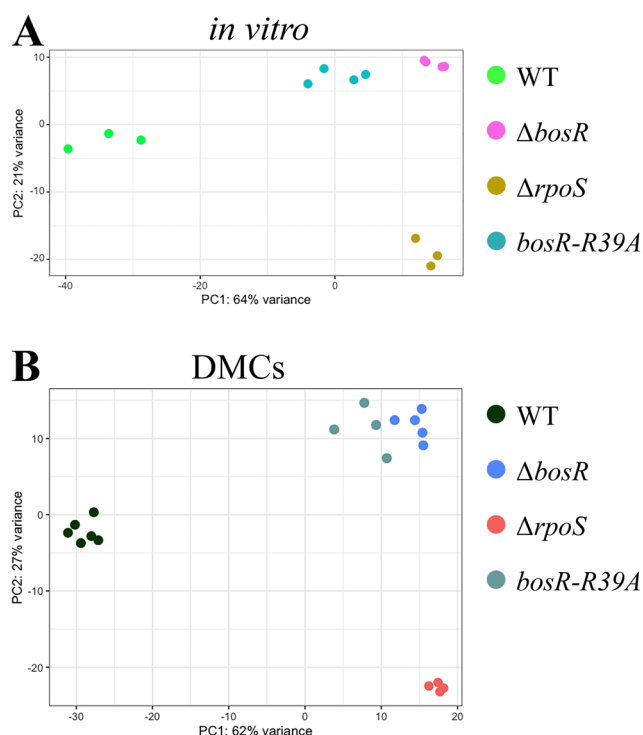
to unambiguously assess the requirement of DNA binding for BosR's RpoS-dependent and -independent functions. We first confirmed that recombinant BosR-R39A no longer binds the *rpoS* promoter using EMSAs (Figure 4A) with a region containing previously identified BosR binding sites (Ouyang et al. 2011, 2015). We then replaced the native *bosR* allele in *B. burgdorferi* B31 5A4 WT strain with *bosR-R39A* and a PflgB-kan kanamycin resistance cassette. Immunoblot analyses of in vitro-cultivated *bosR-R39A* strain revealed that RpoS and OspC were not expressed and DbpA was minimally detected, despite comparable levels of BosR-R39A to BosR in the WT parent strain (Figure 4B, lanes *i* and *ii*). Similarly, immunoblots of DMC-cultivated *bosR-R39A* exhibited undetectable RpoS, minimal OspC and DbpA, and no repression of OspA (Figure 4C, lanes *i* and *ii*, Figure S3). We next performed RNA-seq on in vitro- and DMC-cultivated *bosR-R39A* to assess the global impact of loss of DNA binding. Expression of *rpoS* was significantly reduced in *bosR-R39A* (−13.5-fold in vitro, −6.2-fold in DMCs), resulting in widespread dysregulation of RpoS-dependent genes: 210 of 279 (~75%) in vitro (Table S2) and 138 of 146 (~95%) in DMCs (Table S3). Loss of DNA binding by BosR also led to substantial dysregulation of RpoS-independent genes: 53 of 176 (~30%; 28 activated, 25 repressed) in vitro and 128 of 217 (~59%; 92 repressed, 36 activated) in DMCs (Table 3, Table S3). PCA of in vitro- (Figure 5A) and DMC-cultivated (Figure 5B) samples showed that in both conditions *bosR-R39A* clustered near  $\Delta$ *bosR* and occupied an intermediate position between  $\Delta$ *rpoS* and WT strains in both conditions. This pattern further supported that loss of DNA binding by BosR largely phenocopies loss of BosR itself, resulting in dysregulation of both RpoS-independent and RpoS-dependent genes in vitro and in DMCs.

In our prior work (Grassmann et al. 2023), deleting *bosR* in a  $\Delta$ *rpoS/irpoS* strain harboring an IPTG-inducible *rpoS* allele revealed that BosR is required for proper transcriptional regulation by RNAP-RpoS. Herein, we used a modification of this genetic strategy to confirm that cooperation between BosR and RNAP-RpoS to establish the mammalian-phase RpoS regulon requires DNA binding by BosR. We replaced the native *bosR* allele with *bosR-R39A* in the  $\Delta$ *rpoS/irpoS* background, generating the *bosR-R39A* $\Delta$ *rpoS/irpoS* strain. Upon IPTG induction of *rpoS* in vitro, the *bosR-R39A* $\Delta$ *rpoS/irpoS* strain produced RpoS, OspC, and DbpA in a dose-dependent manner (Figure 4B, lanes *iii–vi*), mirroring the previously demonstrated (Grassmann et al. 2023) response of  $\Delta$ *rpoS/irpoS* (Figure 4B, lanes *vii* and *viii*) and  $\Delta$ *bosR* $\Delta$ *rpoS/irpoS* (Figure 4B, lane *ix*), which are included here as comparators. After DMC cultivation under IPTG-inducing conditions, *bosR-R39A* $\Delta$ *rpoS/irpoS* produced RpoS at WT levels, while OspC and DbpA were diminished and repression of OspA was lost (Figure 4C, lane *iii*, Figure S3). This protein profile closely resembled that of the  $\Delta$ *bosR* $\Delta$ *rpoS/irpoS* strain under the same conditions (Figure 4C, lane *iv*, Figure S3) (Grassmann et al. 2023), reinforcing that loss of DNA binding by BosR largely phenocopies its deletion.

We next performed RNA-seq on IPTG-induced *bosR-R39A* $\Delta$ *rpoS/irpoS* as an alternative strategy to assess the importance of DNA binding by BosR for RNAP-RpoS function. In vitro, 110 of 279 (~39%) RpoS-dependent genes in the BosR regulon were dysregulated in the *bosR-R39A* $\Delta$ *rpoS/irpoS* strain (Table S2). In addition, the degree of dysregulation was



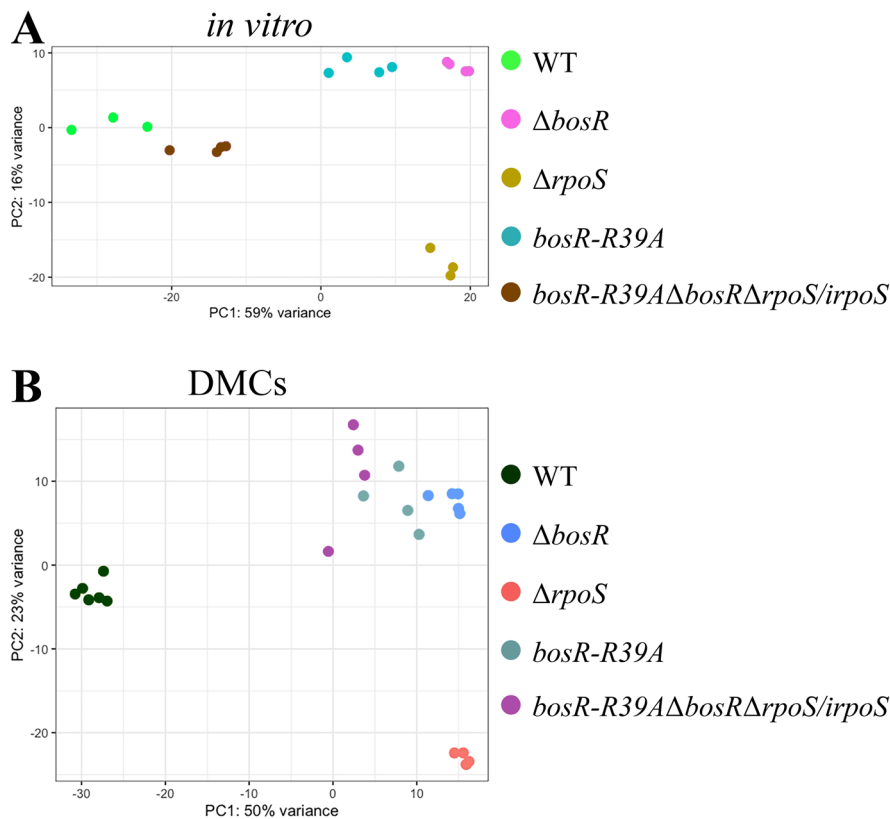
**FIGURE 4** | Expression of *rpoS* and regulation of RpoS-dependent genes require DNA binding by BosR. (A) Electrophoretic mobility shift assay (EMSA) showing that recombinant BosR-R39A fails to bind the *rpoS* promoter region. (B) Immunoblot analysis of WT (lane i), *bosR-R39A* (lane ii), and *bosR-R39AΔrpoS/irpoS* (lanes iii-vi) following in vitro cultivation; the previously generated (Grassmann et al. 2023) *ΔrpoS/irpoS* (lanes vii and viii) and *ΔbosRΔrpoS/irpoS* (lane ix) strains were included as comparators. The IPTG concentration used to induce *rpoS* are indicated above each lane. Note that the lanes were rearranged for clarity; Figure S2 contains the original silver-stained SDS-PAGE and immunoblot images. (C) Immunoblot analysis of DMC-cultivated WT (lane i), *bosR-R39A* (lane ii), and IPTG-induced (+) *bosR-R39AΔrpoS/irpoS* (lane iii), and *ΔbosRΔrpoS/irpoS* (lane iv). Band intensities were quantified by densitometry (Figure S3). Immunoblots are representative of at least three independent experiments performed with a minimum of four biological replicates per strain.



**FIGURE 5** | Loss of DNA binding by BosR disrupts global transcriptional regulation. PCA of WT, *ΔbosR*, *ΔrpoS*, and *bosR-R39A* strain transcriptomes cultivated in vitro (A) and in DMCs (B).

markedly reduced compared to that in the *ΔbosR* mutant; for example, expression of *ospC* in *bosR-R39AΔrpoS/irpoS* was ~5.5-fold lower than in WT but ~1050-fold lower in *ΔbosR* compared to the same WT samples. PCA of in vitro samples showed that *bosR-R39AΔrpoS/irpoS* clustered closest to WT, occupying an intermediate position between WT, *ΔrpoS*, and the cluster formed by *bosR-R39A* and *ΔbosR* (Figure 6A). In contrast, the effect of impaired DNA binding by BosR was more pronounced in DMCs, in which 124 of 146 (~85%; 62 of 73 repressed and 62 of 73 activated) RpoS-regulated genes were dysregulated in *bosR-R39AΔrpoS/irpoS* (Table 2, Table S3). Importantly, tick-phase genes normally repressed by RpoS in host-adapted spirochetes were de-repressed in induced *bosR-R39AΔrpoS/irpoS*. As with in vitro cultivated organisms, the expression of prototypical RpoS-dependent genes was affected to a lesser extent in *bosR-R39AΔrpoS/irpoS*. For instance, *ospC* and *dbpA* transcripts were reduced by 1.75- and 2.95-fold, respectively, in *bosR-R39AΔrpoS/irpoS* relative to WT, compared with 7.5- and 6.45-fold reductions in the *ΔbosR* mutant. Although these differences were smaller at the transcriptional level, they align with the modest decrease in OspC and DbpA protein detected in DMC-cultivated *bosR-R39AΔrpoS/irpoS* compared to WT (Figure 4C, Figure S3) and replicate the phenotype of *bosR* deletion in the inducible *rpoS* strain (*ΔbosRΔrpoS/irpoS*) (Grassmann et al. 2023). PCA of the DMC-cultivated samples further showed that induced *bosR-R39AΔrpoS/irpoS* clustered with *bosR-R39A* and *ΔbosR*,





**FIGURE 6** | Loss of DNA binding by BosR disrupts RpoS-mediated gene regulation in *Borrelia burgdorferi*. PCA of WT,  $\Delta$ *bosR*,  $\Delta$ *rpoS*, *bosR*-R39A, and IPTG-induced *bosR*-R39A $\Delta$ *rpoS*/*irpoS* transcriptomes in strains cultivated in vitro (A) and in DMCs (B).

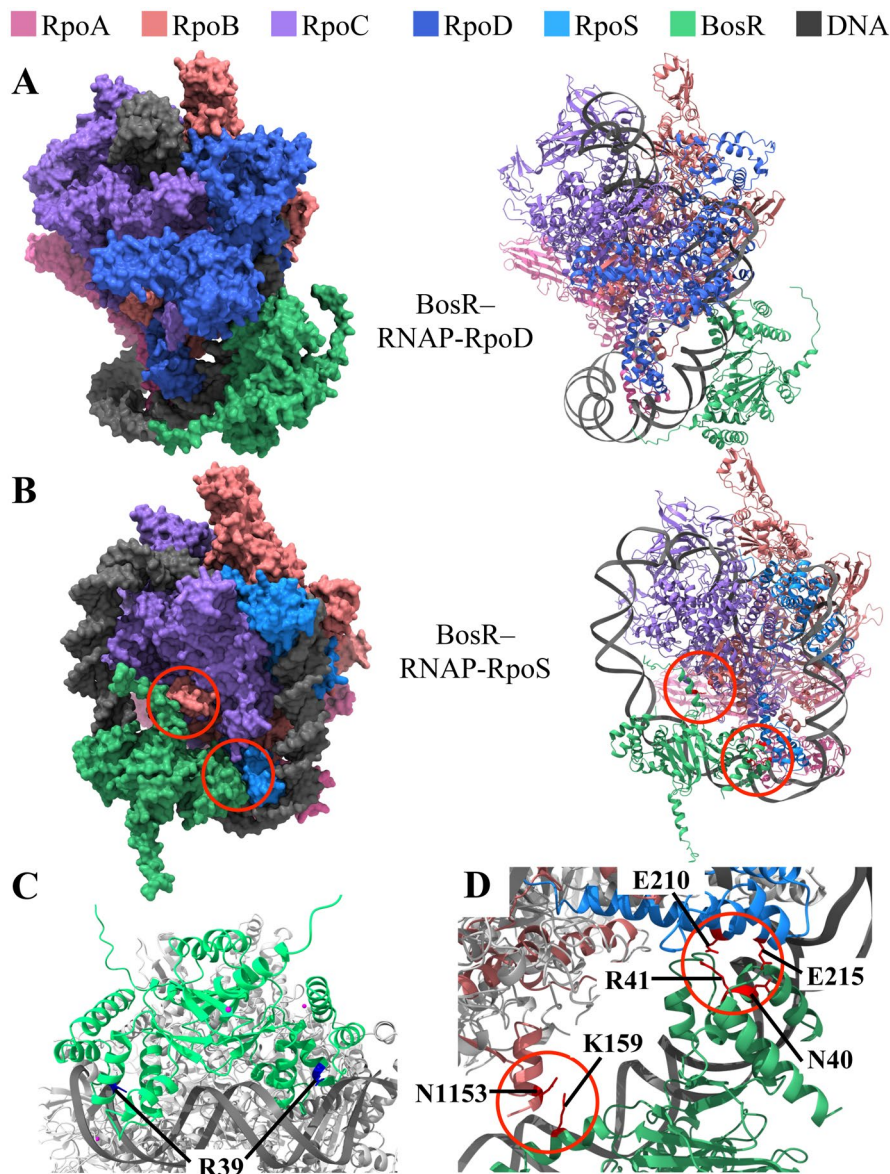
clearly separated from both WT and  $\Delta$ *rpoS* (Figure 6B). A complementary PCA incorporating RNA-seq data from our previously published (Grassmann et al. 2023) DMC transcriptomic analysis of the  $\Delta$ *bosR* $\Delta$ *rpoS*/*irpoS* strain (Figure S4) showed that this strain also clustered with the BosR-deficient mutants. Thus, BosR's DNA-binding activity contributes to RNAP-RpoS-dependent regulation of a subset of genes in vitro but is essential for global modulation of RNAP-RpoS function in vivo.

## 2.4 | BosR Does Not Rely on a Single, Highly Discriminative Consensus DNA Motif

Canonical FUR-family regulators bind conserved DNA motifs known as 'FUR boxes' (Lee and Helmann 2007; Pi and Helmann 2017, 2018; Sarvan, Charih, et al. 2018; Sevilla et al. 2021; Steingard and Helmann 2023), typically 10–20 bp inverted repeats located near regulated promoters. Although prior studies have proposed DNA binding sites for BosR upstream of a small number of genes (Ouyang et al. 2015; Shi et al. 2014; Wang et al. 2013; Sze and Li 2023; Boylan et al. 2003; Katona et al. 2004), these motifs lack a conserved consensus sequence. We reasoned that the in vitro and mammalian-phase BosR regulons defined herein constitute a robust dataset for investigating whether a conserved BosR-binding motif exists. We began by examining two well-validated BosR-binding motifs originally identified upstream of *rpoS* and *ospAB*, each comprising distinct AT-rich

direct repeats (DRs) (Ouyang et al. 2015; Shi et al. 2014). At the *rpoS* promoter region, BosR binds a 10-bp DR sequence, TAAATTAAAT (hereafter *PrpoS*\_DR), and its reverse complement (*PrpoS*\_DR\_rc) (Ouyang et al. 2015). These DRs also were confirmed as BosR targets at the *cheW2* promoter (Sze and Li 2023). BosR binding upstream of *ospAB* involves two 14-bp DRs, AACCAAACTTAATT (*PospA*\_DR1/2) (Shi et al. 2014). To assess the prevalence of these motifs genome-wide, we used FIMO (MEME suite) (Grant et al. 2011; Bailey et al. 2015) to search both strands of the 150 bp upstream regions of all genes in the in vitro and mammalian-phase BosR regulons (based on either mapped transcription start sites (Adams et al. 2017) or, when unavailable, start codons). Approximately 25% of genes in the in vitro BosR regulon and 28% in the mammalian-phase regulon contained at least one of these motifs (Table S5). However, motif frequencies were nearly identical among genes outside the BosR regulons (~27% in vitro, ~26% in DMCs). To refine this analysis, we aligned all instances of *PrpoS*\_DR, *PrpoS*\_DR\_rc and *PospA*\_DR1/2 located upstream of BosR-regulated genes, which yielded the AT-rich consensus sequence ATTTTAAATTAAAT (Figure S5). Expanding our FIMO search to include variants of this consensus motif and its reverse complement increased detection to ~45% of genes in the in vitro BosR regulon and ~50% in the mammalian-phase regulon. However, these variants were also present at virtually identical frequencies (~48%) among non-regulated genes. Finally, we used MEME (Bailey and Elkan 1994; Bailey et al. 2015) to perform *de novo* motif discovery within the same 150 bp regions. Even under relaxed





**FIGURE 7** | Structural modeling supports a bifunctional model for BosR: a canonical FUR-like regulator with RNAP-RpoD and a sigma tether with RNAP-RpoS. (A) AlphaFold3-predicted structure of BosR and the RNAP-RpoD complex in which BosR does not interact with RNAP-RpoD. Left: surface rendering; right: cartoon representation. (B) Predicted structure of the BosR-RNAP-RpoS complex. Left: surface rendering; right: cartoon representation. Predicted contact sites between BosR and RNAP-RpoS are highlighted with red circles. (C) Close-up of B with BosR bound to DNA, highlighting residue R39 (blue). (D) Two interfaces predicted between BosR and RNAP-RpoS: (i) K159 of BosR with N1153 of the RpoB subunit; (ii) N40/R41 of BosR with E215/E210 in region 4 of RpoS.

stringency, MEME failed to identify a statistically significant motif. Taken together, these findings suggest that BosR does not rely on a single, highly discriminative consensus sequence motif to direct transcriptional regulation.

## 2.5 | Structural Modeling Predicts That BosR Binds DNA and Functions as a Sigma Tether at RpoS-Regulated Genes

Our collective data suggests that BosR functions as a canonical FUR at RpoS-independent genes within the BosR regulons but acts in a noncanonical manner at RpoS-regulated promoters by engaging RNAP-RpoS while also binding DNA. To explore these possibilities at the structural level, we used AlphaFold3 (AF3)

(Abramson et al. 2024) to generate models of BosR with RNAP-RpoD and RNAP-RpoS. Although AF3 does not predict DNA sequence-specific interactions, it can incorporate DNA backbone structures to provide contextual positioning (Abramson et al. 2024). We therefore included random DNA sequences in each model to simulate promoter-bound complexes. In a representative high-confidence model with BosR and RNAP-RpoD (Figure 7A), BosR does not contact the holoenzyme. In contrast, a high-quality model of the BosR-RNAP-RpoS complex predicted that BosR simultaneously engages RNAP-RpoS (Figure 7B) and DNA (Figure 7C) with two distinct interfaces predicted between BosR and RNAP-RpoS (Figure 7D): (i) K159 in BosR's C-terminus with N1153 in the RNAP  $\beta$ -subunit and (ii) N40/R41 in BosR's recognition helix with E215/E210 in region 4 of RpoS.

### 3 | Discussion

The FUR family encompasses a diverse group of transcription factors that typically sense either intracellular metal availability—such as iron in Fur, zinc in Zur, or nickel in Nur—through direct binding of the cognate metal to a regulatory site, or, in the case of PerR, hydrogen peroxide via metal-catalyzed oxidation of key histidine residues (Lee and Helmann 2007; Sarvan, Butcher, et al. 2018). In canonical FURs, ligand binding triggers a conformational change that enables high-affinity DNA binding, allowing the regulator to repress genes involved in metal uptake and utilization or oxidative stress response (Sarvan, Butcher, et al. 2018; Sarvan, Chari, et al. 2018; Sevilla et al. 2021). In recent years, accumulating evidence has challenged this conventional view, revealing that many FUR proteins also serve as transcriptional activators and exhibit diverse modes of DNA recognition in both the regulatory metal-bound and -unbound states (Steingard and Helmann 2023; Gao et al. 2023; Sevilla et al. 2021; Sarvan, Butcher, et al. 2018). Notably, FUR family members have been shown to regulate virulence genes in diverse bacterial pathogens, positioning them as global regulators that integrate metal sensing with broader aspects of pathogenesis (Steingard and Helmann 2023; Grassmann et al. 2021; Delany et al. 2004; Brenot et al. 2005; Carpenter et al. 2009; Rea et al. 2004; Gao et al. 2023). BosR stands out as an extreme example of this expanded paradigm of FURs as highly versatile transcriptional regulators (Steingard and Helmann 2023). While early studies linked BosR to oxidative stress responses (Hyde et al. 2010; Boylan et al. 2003; Seshu et al. 2004; Hyde et al. 2009, 2006; Esteve-Gassent et al. 2009), the discovery that it activates transcription of *rpoS* (Hyde et al. 2009, 2010; Ouyang et al. 2011, 2009, 2015; Katona 2015) and regulates the RpoN/RpoS signaling cascade (Grassmann et al. 2023) underscored BosR as an atypical FUR. Our work builds upon these observations by showing that, in addition to regulating RpoS-dependent gene expression in concert with RNAP-RpoS (Grassmann et al. 2023), BosR also regulates a separate cohort of RpoS-independent genes in host-adapted *B. burgdorferi*. Importantly, these regulatory activities are most consistent with DNA binding by BosR, which appears to occur largely in the absence of a conserved consensus motif and may be broadly permissive in sequence specificity. Unlike canonical FURs, BosR lacks the regulatory metal-binding site typically required for sensing either metal availability or hydrogen peroxide, suggesting that it may adopt a conformation that is inherently competent for DNA binding. BosR's regulatory plasticity is likely an evolutionary adaptation to *B. burgdorferi*'s streamlined genome and atypical metal usage, particularly its minimal reliance on iron (Posey and Gherardini 2000).

The absence of a canonical peroxide-sensing metal-binding site in BosR has raised long-standing questions (Boylan et al. 2003; Seshu et al. 2004; Hyde et al. 2006; Wang et al. 2017) about how it might detect oxidative stress and regulate genes such as *napA/dps*, *sodA*, and *cdr*. Collectively, as suggested by prior in vitro studies (Hyde et al. 2009; Ouyang et al. 2009, 2011), our in vitro and in vivo transcriptomic analyses herein provided an anticipated solution to this conundrum: BosR does not regulate these key enzymatic components of *B. burgdorferi*'s oxidative stress response. Instead, it appears to modulate a distinct set of RpoS-independent genes involved in diverse cellular functions,

some of which, such as *uvrD* and *mutS*, are linked to the repair of DNA damage induced by oxidative/nitrosative stress (Bourret et al. 2016). BosR's role in defense against oxidative/nitrosative stress, therefore, is likely more nuanced and may be tailored to specific phases of the enzootic cycle, such as the tick midgut during the blood meal, where these challenges are more likely to be relevant (Bourret et al. 2016, 2019). Although the nucleotide excision repair (NER) system in *B. burgdorferi*, which includes *uvrD*, has been investigated primarily in the context of the ROS/RNS response (Bourret et al. 2011; Hardy and Chaconas 2013; Troxell et al. 2014; Bourret et al. 2016; Ramsey et al. 2017), in other bacteria it also contributes to replication fork stability and transcription-coupled repair, thereby preserving genome integrity under replication stress or transcriptional blockages (Wozniak and Simmons 2022; Kraithong et al. 2021; Epshtein et al. 2014). Similarly, *mutS*, a key component of the mismatch repair (MMR) pathway (Marinus 2012), plays broader roles in other bacteria, including genome surveillance, suppression of recombination between divergent sequences, and modulation of mutation rates in response to environmental stressors (Reyes et al. 2015; Tham et al. 2013; Marinus 2012). Consistent with a genome maintenance role, BosR also regulates several genes involved in plasmid or chromosome partitioning, which in *B. burgdorferi* is essential for faithful segregation of its linear chromosome and complex assortment of plasmids during cell division (Ren et al. 2023; Takacs et al. 2022; Casjens et al. 2000). Beyond DNA repair and oxidative/nitrosative stress responses, our transcriptomic analysis of DMC-cultivated organisms revealed that some BosR-repressed genes, such as *cspZ*, *pncA*, and purine permeases, are essential for mammalian infection (Hartmann et al. 2006; Lin et al. 2020; Jain et al. 2012; Purser et al. 2003); these results indicate that BosR may fine-tune the expression of these virulence-associated genes to prevent potentially detrimental overexpression. Conceivably, this regulatory balance is coordinated with other transcription factors, including SpoVG and BadR, both of which are regulated by BosR independently of RpoS in host-adapted spirochetes (Grassmann et al. 2023). BadR is particularly relevant given its direct binding to the *bosR* and *rpoS* promoters (Ouyang and Zhou 2015; Miller et al. 2013; George and Ouyang 2025). Notably, although the BadR regulon was defined under in vitro conditions (George and Ouyang 2025), it overlaps substantially with the mammalian-phase BosR regulon, suggesting potential convergence of their regulatory networks in vivo. These findings raise the possibility that BosR and BadR reciprocally regulate *rpoS* and potentially each other, forming a regulatory circuit that integrates into the broader RpoN/RpoS pathway and influences a significant portion of the BosR regulon.

The presumptive ability of BosR to regulate distinct regulons—one in vitro and another in vivo—without a unifying DNA motif further redefines the boundaries of FUR-family function. BosR has been shown to bind with high affinity to discrete motifs upstream of several genes (Ouyang et al. 2015; Shi et al. 2014; Wang et al. 2013; Sze and Li 2023; Boylan et al. 2003; Katona et al. 2004), including *rpoS* (Ouyang et al. 2015) and *ospAB* (Shi et al. 2014). However, our bioinformatics analyses revealed no single, highly discriminative consensus motif shared across the BosR regulons. Experimental confirmation of these findings will require genome-wide mapping of BosR–DNA interactions

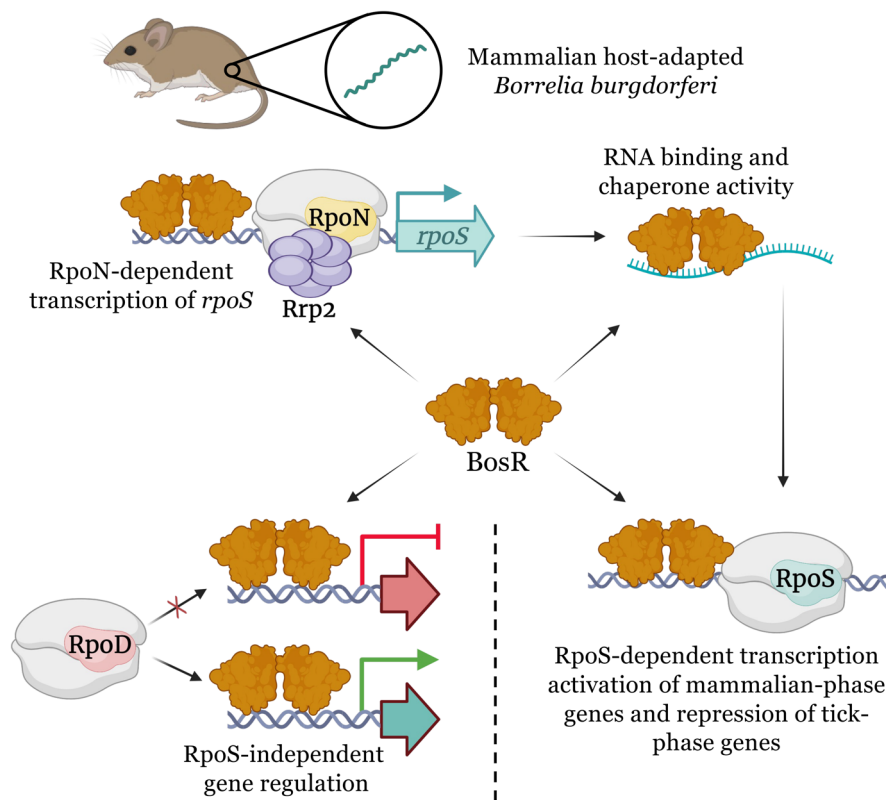


(e.g., chromatin immunoprecipitation followed by sequencing, ChIP-seq) to define its binding landscape in vivo. Notably, the apparent absence of a consensus BosR-binding motif aligns with growing evidence that FURs, traditionally viewed as sequence-specific regulators that recognize conserved 'FUR boxes' (Lee and Helmann 2007), also can modulate transcription through interactions with DNA lacking a consensus motif (Sevilla et al. 2021; Gao et al. 2023; Pi and Helmann 2018; Gilston et al. 2014). BosR appears to regulate gene expression through high-affinity, sequence-specific binding at a limited subset of genes, and through lower-affinity, sequence-permissive interactions for the majority of its regulon. Similar binding strategies are employed by other global regulators, particularly nucleoid-associated proteins (NAPs) (Dorman et al. 2020; Bouffartigues et al. 2007; Lang et al. 2007) such as FIS, H-NS, and IHF, and even CRP (Grainger et al. 2005; Dorman et al. 2020), which use consensus sites as anchors while also engaging DNA non-specifically through electrostatic contacts or sensing of DNA topology. Notably, *B. burgdorferi* encodes other DNA-binding proteins with varying levels of sequence-non-specific binding properties, such as Hbb (Kobryn et al. 2000; Mouw and Rice 2007; Tilly et al. 1996), SpoVG (Saylor et al. 2023; Savage et al. 2018) and members of the Pfm12 family (BBK01, BBG01, BBH37, BBJ08, and BB0844) (Brangulis et al. 2024). These proteins could participate in shaping the local DNA environment or creating higher-order chromatin-like structures, raising the possibility that BosR-dependent transcriptional control is either influenced by, or contributes to, global DNA architecture. Consistent with this idea, the composition of the BosR regulon and its DNA-binding requirements differ markedly between in vitro and in vivo growth conditions, suggesting that environmental context alters DNA recognition, promoter occupancy, and/or accessibility. The RpoS regulon is likewise dynamically remodeled within different environmental conditions, including in vitro cultivation, fed nymphs, and mammalian host adaptation (Caimano et al. 2007, 2019; Grassmann et al. 2023), a phenomenon that also may reflect differences in DNA topology and chromatin structure. Several transcription regulators are differentially expressed by *B. burgdorferi* in vitro and in vivo (Samuels et al. 2021), which could change the local DNA context, altering how BosR and RpoS bind DNA at different promoters. Finally, our finding that certain genes in the in vitro and in vivo BosR regulons are regulated normally even without BosR DNA binding indicates that some transcriptional control is indirect, potentially involving other BosR-regulated DNA-binding proteins (e.g., BadR, SpoVG) or uncharacterized small RNAs (Lybecker and Samuels 2017; Medina-Perez et al. 2020; Petroni et al. 2023), adding further complexity to the regulatory network.

By deleting *bosR* in the inducible *rpoS* strain ( $\Delta$ *bosR* $\Delta$ *rpoS*/*irpoS*), we recently demonstrated that BosR is essential for proper RNAP-RpoS function in mammalian host-adapted *B. burgdorferi* (Grassmann et al. 2023). Although earlier in vitro studies identified BosR as a critical factor for silencing RpoS-repressed genes such as *ospAB* (Wang et al. 2013; Shi et al. 2014), our current and prior (Grassmann et al. 2023) work with host-adapted spirochetes revealed that BosR is required not only for RpoS-mediated repression of tick-phase genes but also for activation of RpoS-dependent transcription during mammalian adaptation. Here, we show that RpoS-mediated gene regulation is nearly

abolished in the induced *bosR-R39A* $\Delta$ *rpoS*/*irpoS* strain, supporting the contention that DNA binding by BosR is an essential determinant of the RpoS regulon. Furthermore, the requirement of BosR DNA binding for repression suggests that BosR may stabilize RNAP-RpoS at target promoters, thereby excluding RNAP-RpoD and establishing non-productive complexes. This model aligns with our prior promoter mapping studies demonstrating that minimal  $-10/-35$  elements are sufficient for RpoS-dependent repression of tick-phase genes such as *ospA* and *glp* (Grove et al. 2017). The apparent absence of a conserved BosR-binding motif upstream of these genes, combined with the observation that previously proposed BosR boxes are A/T-rich (Ouyang et al. 2015; Shi et al. 2014; Wang et al. 2013; Sze and Li 2023; Boylan et al. 2003; Katona et al. 2004), suggests that BosR may bind DNA in a sequence-permissive manner, favoring open A/T-rich complexes. We therefore propose that direct interaction between BosR and RNAP-RpoS, along with promoter engagement by the sigma factor, dictates BosR's specificity for RpoS-regulated promoters. In this capacity, BosR functions analogously to sigma factor activators—a growing class of structurally diverse and evolutionarily unrelated proteins, including Crl in *Escherichia coli* (Cartagena et al. 2019; Xu et al. 2019), RbpA in Actinobacteria (Hubin et al. 2017, 2015; Tabib-Salazar et al. 2013), GrgA in *Chlamydia trachomatis* (Bao et al. 2012; Desai et al. 2018), and GcrA in *Caulobacter crescentus* (Wu et al. 2023)—that promote transcription by binding both sigma factors and promoter DNA (Vishwakarma and Brodolin 2020). Like BosR, these "sigma tethers" often interact with DNA in a sequence-nonspecific manner (Cartagena et al. 2019; Xu et al. 2019; Vishwakarma and Brodolin 2020; Bao et al. 2012; Desai et al. 2018). Although canonical FUR proteins regulate transcription via DNA binding without directly contacting RNAP holoenzymes (Lee and Helmann 2007; Sarvan, Butcher, et al. 2018; Sevilla et al. 2021; Steingard and Helmann 2023; Gilston et al. 2014; Kang et al. 2024; Liu et al. 2021; Sarvan, Charhi, et al. 2018), a direct interaction between a FUR-family regulator and RNAP has been described (Yang et al. 2022). The proposed mechanism in which BosR engages both RNAP-RpoS and promoter DNA thus represents a striking repurposing of FUR-family mechanisms. This mode of action may also explain the toxicity associated with RpoS overproduction in in vitro-cultivated *B. burgdorferi* (Chen et al. 2013; Grassmann et al. 2023; Wachter et al. 2023), which is alleviated by deletion of *bosR* (Grassmann et al. 2023). Excess RpoS likely drives abnormal accumulation of BosR–RNAP–RpoS complexes, depleting the pool of RNAP apoenzyme available for complex formation with RpoD. The alleviation of RpoS toxicity in the *bosR-R39A* $\Delta$ *rpoS*/*irpoS* background (data not shown) further supports BosR's role as a sigma factor activator.

Recent studies suggest that BosR may bind not only DNA but also RNA, an activity that appears to be unique among FUR-family regulators, none of which are known to function primarily as RNA-binding or RNA-chaperone proteins. Raghunandan et al. (2024) proposed that, rather than engaging the *rpoS* promoter directly, BosR interacts with the 5'-UTR of the *rpoS* mRNA, stabilizing the transcript and thereby enhancing RpoS production via a post-transcriptional mechanism. Although elegantly presented, this model conflicts with extensive evidence from multiple laboratories (Ouyang et al. 2011; Wang et al. 2013; Shi



**FIGURE 8** | Model of BosR-mediated gene regulation in *B. burgdorferi* during mammalian host infection. BosR requires DNA binding to activate RpoN-dependent transcription of *rpoS* (top left). BosR may also modulate *rpoS* and other transcripts post-transcriptionally through an additional mechanism that involves interactions with RNA (top right). At RpoS-regulated promoters, BosR binds DNA and likely engages RNA polymerase holoenzyme containing RpoS, functioning as a sigma factor activator to drive transcription of mammalian-phase genes and repression of tick-phase genes (bottom right). At RpoD-dependent promoters, BosR functions as a canonical FUR-like regulator, binding upstream elements to activate or repress target genes independently of RpoS (bottom left).

et al. 2014; Katona 2015; Ouyang et al. 2015; Mason et al. 2019), including the data presented here, showing that RpoN-dependent transcription of *rpoS* requires high-affinity DNA binding by BosR. In canonical FUR-family regulators, a conserved arginine in the helix–turn–helix DNA-binding domain anchors the protein to DNA through electrostatic and hydrogen-bond interactions (Sarvan, Butcher, et al. 2018; Sarvan, Charih, et al. 2018; Sevilla et al. 2021; Deng et al. 2015; Yang et al. 2022; Kang et al. 2024; Gilston et al. 2014; Lin et al. 2014). BosR retains this equivalent residue (R39), which our data demonstrate is indispensable for BosR-dependent transcriptional regulation. Nonetheless, we cannot entirely exclude the possibility that R39 also contributes to RNA interactions post-transcriptionally. More recently, Van Gundy et al. (2025) identified BosR among a group of *B. burgdorferi* proteins exhibiting detectable RNA-binding or RNA-chaperone activity in vitro. The biological relevance of this activity during the enzootic cycle and how it relates to regulation of RpoS-dependent and -independent genes by BosR remains to be determined.

Taken together, our findings support a functional model (Figure 8) in which BosR regulates transcription through two distinct mechanisms. At RpoS-regulated promoters, BosR acts as a sigma factor activator—or “sigma tether”—that facilitates engagement of RNAP-RpoS via both protein–protein and protein–DNA interactions. At RpoD-dependent, RpoS-independent genes, by contrast, BosR functions more

conventionally as a FUR-like transcription factor, binding DNA directly without interacting with the RNAP holoenzyme. This dual functionality helps explain BosR's broad regulatory scope and its central role in mammalian host adaptation. BosR may also modulate expression of *rpoS* and other genes post-transcriptionally through a poorly understood mechanism that may involve interactions with RNA. Our findings further suggest that BosR intersects with multiple regulatory circuits that govern gene expression in *B. burgdorferi*. BosR is expressed throughout the enzootic cycle (Ouyang et al. 2016; Medrano et al. 2007; Grassmann et al. 2023), including during larval acquisition, when RpoS is OFF (Caimano et al. 2007; Dunham-Ems et al. 2012; Ouyang et al. 2012; Caimano et al. 2019), indicating a role for BosR in controlling a distinct, RpoS-independent regulon in ticks. Notably, we have shown previously that the tick-phase-specific second messenger c-di-GMP, acting through its receptor PlzA, modulates BosR's activities with RNAP-RpoS during transmission (Grassmann et al. 2023). This observation suggests that liganded-PlzA and/or other tick-specific factors could also influence BosR's RpoS-independent functions in ticks. The marked differences between the BosR regulons defined in vitro and in vivo raise the intriguing possibility that the in vitro regulon may, at least in part, reflect a tick-phase gene expression program. Understanding how environmental cues and host-specific signals modulate BosR activity during different phases of the enzootic cycle will be key to defining its full regulatory capacity.



## 4 | Experimental Procedures

### 4.1 | Ethics Statement

All animal experiments conducted at UConn Health complied with the Guide for the Care and Use of Laboratory Animals (8th Edition) and were approved by the UConn Health Institutional Animal Care and Use Committee (Animal Welfare Assurance number A347-01).

### 4.2 | Bacterial Strains and Cultivation of *B. burgdorferi* In Vitro and in DMCs

*Escherichia coli* strains TOP10 (ThermoFisher Scientific, Waltham, MA) and Stellar (Takara Bio USA, San Jose, CA) were used for cloning and plasmid propagation and maintained in Lysogeny Broth (LB) or on LB agar supplemented with the appropriate antibiotics: ampicillin ( $100\mu\text{g mL}^{-1}$ ), spectinomycin ( $100\mu\text{g mL}^{-1}$ ), kanamycin ( $100\mu\text{g mL}^{-1}$ ), and/or gentamicin ( $5\mu\text{g mL}^{-1}$ ). *B. burgdorferi* strains (Table S6) were cultivated in Barbour–Stoenner–Kelly (BSK)-II medium supplemented with 6% rabbit serum (Pel-Freeze Biologicals, Rogers, AR) and antibiotics when appropriate: kanamycin ( $400\mu\text{g/mL}$ ), streptomycin ( $100\mu\text{g mL}^{-1}$ ), and gentamicin ( $50\mu\text{g mL}^{-1}$ ). *B. burgdorferi* plasmid contents were verified by PCR as described previously (Bunikis et al. 2011; Groshong, McLain, and Radolf 2021). For IPTG-inducible *rpoS* (*irpoS*) strains, cultures were grown in BSK-II supplemented with the appropriate antibiotics and IPTG at concentrations ranging from 0.01 to 1.0 mM, as described previously (Grassmann et al. 2023). Growth curves for WT and  $\Delta\text{bosR}$  were performed in quadruplicate by inoculating cultures at  $1 \times 10^4$  spirochetes/mL in BSK-II containing the appropriate antibiotics and incubating at  $37^\circ\text{C}$ . Growth curves were compared using the CGGC permutation test (Elso et al. 2004) with 1000 permutations. Spirochete densities were enumerated daily by darkfield microscopy using a Petroff–Hausser counting chamber. For in vitro RNA-seq experiments, spirochetes grown to mid-logarithmic phase at  $23^\circ\text{C}$  were used to inoculate fresh cultures at  $1 \times 10^4$  spirochetes/mL. These were then shifted to  $37^\circ\text{C}$  in 5%  $\text{CO}_2$  and 1%  $\text{O}_2$  and harvested at late-logarithmic to early stationary phase.  $37^\circ\text{C}$  in vitro cultures of *bosR-R39A* $\Delta\text{rpoS/irpoS}$  for RNA-seq analyses were supplemented with 1.0 mM IPTG. For mammalian host-adapted spirochetes, *B. burgdorferi* was cultivated within DMCs (8–14 kDa MWCO; Spectra/Por, Spectrum Laboratories) implanted into the peritoneal cavities of Sprague–Dawley rats (Envigo RMS Inc., Indianapolis, IN) for 13 days as described previously (Caimano 2018; Grassmann et al. 2023). IPTG-induction of *rpoS* during mammalian host adaptation was achieved by replacing untreated rat drinking water with water containing 2% sucrose and 80 mM IPTG beginning at least 7 days prior to DMC implantation and continuing throughout the 13-day cultivation period, as previously described (Grassmann et al. 2023).

### 4.3 | Routine DNA Manipulation and Cloning

Plasmids were purified from *E. coli* using QIAprep spin, midi or mega kits (Qiagen, Germantown, MD). Bacterial genomic DNA was extracted using the Gentra Puregene Yeast/Bacteria kit (Qiagen). Oligonucleotide primers used in these studies were purchased from Sigma-Aldrich (St. Louis, MO).

Cloning was performed using the In-Fusion HD Cloning Plus kit (TaKaRa Bio USA Inc.). Routine and high-fidelity PCR amplifications were performed using RedTaq (Denville Scientific, Holliston, MA) and CloneAmp HiFi (TaKaRa Bio USA Inc.), respectively. Sanger sequencing of cloned DNAs was performed by Genewiz Inc. (South Plainfield, NJ) and analyzed using MacVector (MacVector Inc., Apex, NC). *B. burgdorferi* strains were transformed by electroporation as previously described (Samuels 1995).

### 4.4 | Construction of *B. burgdorferi* Mutant Strains

The  $\Delta\text{bosR}$  mutant (BbAG675) was generated in the B31 5A4 background by allelic exchange, replacing *bosR* (*bb0647*) with a *PflgB-kanR* cassette carried on the previously generated plasmid pMC5115 (Grassmann et al. 2023). The *bosR-R39A* point mutant allele was generated by site-directed mutagenesis using the In-Fusion HD Cloning Plus Kit (Takara Bio USA Inc.) in the plasmid pMC5672 (Table S6) and introduced into the *bosR* locus of B31 5A4 to yield the *bosR-R39A* (BbAG690) strain. The *bosR-R39A* allele was subsequently introduced into the IPTG-inducible  $\Delta\text{rpoS/irpoS}$  background (BbAG351) (Grassmann et al. 2023) to generate *bosR-R39A* $\Delta\text{rpoS/irpoS}$  (BbAG692). All mutant strains were verified by PCR, Sanger sequencing, and immunoblotting.

### 4.5 | Recombinant Protein Production, SDS-PAGE, and Immunoblotting

His-tagged FlaB, RpoS, OspC, DbpA, OspA, BosR, and BosR-R39A were overexpressed in *E. coli* Rosetta 2 (DE3) pLysS (MilliporeSigma, Burlington, MA) and purified by nickel affinity chromatography using HisTrap HP columns (Cytiva, Marlborough, MA), followed by size-exclusion chromatography on a Superdex 200 10/300 GL column (Cytiva). Protein purity was assessed by SDS-PAGE with Coomassie Brilliant Blue staining (Thermo Fisher Scientific) and/or by immunoblotting. Recombinant BosR and BosR-R39A were used in EMSAs as described below. Purified recombinant BosR, FlaB, RpoS, OspC, DbpA, and OspA (40–60  $\mu\text{g}$ ) were used to generate polyclonal antisera in female Sprague–Dawley rats (Envigo, South Easton, MA) as previously described (Grassmann et al. 2021). Whole-cell lysates were prepared from *B. burgdorferi* cultures by pelleting approximately  $2 \times 10^8$  spirochetes cultivated either in vitro or in DMCs, washing  $3 \times$  in PBS, and resuspending in Laemmli sample buffer. Proteins were resolved by 12.5% SDS-PAGE, followed by silver staining or transfer to nitrocellulose membranes using a Trans-Blot SD semi-dry apparatus (Bio-Rad, Hercules, CA). Membranes were blocked for 1 h at room temperature in blocking solution containing 5% nonfat dry milk, 0.1% Tween 20, and 5% fetal calf serum in PBS, then probed with rat antisera. Blots were incubated with primary antibodies overnight at  $4^\circ\text{C}$ , washed  $5 \times$  in PBS-Tween, and developed using HRP-conjugated secondary antibodies (Southern Biotechnology Associates) and Pierce SuperSignal West Pico chemiluminescent substrate (Thermo Fisher Scientific). Band intensities from immunoblots were quantified using Gel Analyzer (v. 2020.1, Istvan Lazar) by measuring the integrated density of each band, normalizing signal intensities to the FlaB loading control, and expressing values relative to WT samples.

#### 4.6 | Electrophoretic Mobility Shift Assays (EMSAs)

EMSAs were performed using the LightShift Chemiluminescent EMSA kit (ThermoFisher Scientific) according to the manufacturer's instructions. Briefly, 20  $\mu$ L reactions contained biotin end-labeled DNA fragments encompassing the *rpoS* promoter region (Ouyang et al. 2011, 2015), 0–2  $\mu$ M recombinant BosR or BosR-R39A, and EMSA binding buffer (20 mM HEPES, 5% glycerol, 1 mM DTT, 100  $\mu$ g mL<sup>-1</sup> BSA, 1 mM MgCl<sub>2</sub>, and 50 mM KCl, pH 7.5). After 20 min incubation at room temperature, samples were resolved on 8% native polyacrylamide gels in 0.5× TBE (45 mM Tris-borate, 1 mM EDTA, pH 8.0) at 200 V and transferred to nylon membranes. Biotin-labeled DNA was detected using streptavidin–HRP conjugate and chemiluminescent substrate.

#### 4.7 | RNA Isolation, Library Preparation, and RNA-Seq

Total RNA was isolated from in vitro- or DMC-cultivated *B. burgdorferi* (six biological replicates per strain and condition were initially prepared) using TRIzol (ThermoFisher Scientific) followed by two treatments with TURBO DNase (ThermoFisher Scientific) and purification with RNeasy columns (Qiagen) as previously described (Grassmann et al. 2021, 2023). RNA integrity was verified using a TapeStation 4200 (Agilent Technologies) and only samples with a RNA Integrity Number (RIN)  $\geq 7$  were carried forward. Because a subset did not meet this criterion, the final dataset included a variable number of high-quality biological replicates (minimum of three per strain and condition). Library construction was performed by the University of Connecticut's Center for Genome Innovation (CGI) using the Illumina Stranded Total RNA Ligation kit with Ribo-Zero rRNA depletion. Libraries were validated for size distribution, quantified using a Qubit dsDNA HS Assay Kit (ThermoFisher Scientific), and screened on an Illumina MiSeq (Nano v2, 300-cycle kit) prior to high-throughput sequencing. Sequencing was performed on either the Illumina NextSeq 550 or the NovaSeq 6000 platform using the High Output v2.5 75-cycle kit. Reads were quality-trimmed with Sickle (v1.33) (Joshi 2011), mapped to our custom *B. burgdorferi* B31 genome (Grassmann et al. 2023) using EDGE-pro (v1.1.3) (Magoc et al. 2013), and analyzed for differential expression with DESeq2 (v1.26.0) (Love et al. 2014). Differentially expressed genes were defined as  $\geq 2$ -fold change with Benjamini–Hochberg-adjusted *p* values (*q*)  $< 0.05$ . PCA plots were generated in RStudio using gplots, ggplot2, gtools, and pheatmap. Raw data are deposited in NCBI SRA under accession PRJNA1307481 (Table S1).

#### 4.8 | DNA-Binding Motif Search

Promoter sequences for BosR-regulated genes were retrieved as the 150 bp upstream of the transcription start site (TSS), when available, from a previously published *B. burgdorferi* TSS dataset (Adams et al. 2017); orphan and antisense TSSs were excluded. For genes without a defined TSS, the 150 bp upstream of the start codon was used. We first compiled sequences for four experimentally supported BosR-binding motifs: the direct repeat (DR) in

the *rpoS* promoter (*PrpoS\_DR*), its reverse complement (*PrpoS\_DR\_rc*) (Ouyang et al. 2015), and the two DRs in the *ospAB* promoter (*PospA\_DR1* and *PospA\_DR2*) (Shi et al. 2014). These motifs were scanned against both strands of the promoter sequences for all genes in the in vitro and mammalian-phase BosR regulons using FIMO (v5.5.8, MEME suite) (Grant et al. 2011; Bailey et al. 2015) with a *p*-value threshold of 0.0001 and the default background model. Identical searches were performed for genes outside the BosR regulon to assess background motif frequencies. All instances of *PrpoS\_DR*, *PrpoS\_DR\_rc*, and *PospA\_DR1/2* located upstream of BosR-regulated genes were aligned to generate a consensus sequence (ATTTTAAATTAAAT). This consensus motif and its reverse complement were then used in additional FIMO searches. Sequence logos were generated using WebLogo (v3.7.12) after extending motif instances with flanking genomic sequence. Finally, *de novo* motif discovery was performed on the same promoter sequences using MEME (v5.5.8) (Bailey and Elkan 1994; Bailey et al. 2015) with a site distribution of zero or one occurrence per sequence (ZOOPS) and minimum/maximum widths of 6–20 bp, under relaxed statistical thresholds, to identify potential novel BosR-binding motifs.

#### 4.9 | Bioinformatics and Structural Modeling

Conserved domain searches and protein annotation were performed using UniProt (UniProt 2023) and InterPro (Blum et al. 2021). Multiple sequence alignments were generated with Clustal Omega (Sievers et al. 2011). AlphaFold3 (Abramson et al. 2024) was used to model *B. burgdorferi* (strain B31) BosR and RNA polymerase holoenzymes containing either RpoD or RpoS, using amino acid sequences for RpoA, RpoB ( $\times 2$ ), RpoC, RpoZ, and the appropriate sigma factor. Models were generated both in the absence and presence of duplex DNA fragments to visualize potential sigma factor-dependent conformational differences. Structural visualizations and image rendering were performed with the PyMOL Molecular Graphics System v2.3.2 (Schrödinger LLC, New York, NY) and ChimeraX (Pettersen et al. 2021).

#### Author Contributions

**André A. Grassmann:** conceptualization, investigation, writing – original draft, funding acquisition, methodology, validation, visualization, writing – review and editing, formal analysis, supervision, project administration, data curation. **Melissa A. McLain:** investigation, methodology, validation, writing – review and editing. **Michael R. Freeman:** conceptualization, investigation, methodology, validation, visualization, writing – review and editing. **Melissa J. Caimano:** conceptualization, funding acquisition, methodology, formal analysis, project administration, investigation, writing – review and editing. **Justin D. Radolf:** conceptualization, investigation, funding acquisition, writing – original draft, writing – review and editing, visualization, validation, methodology, formal analysis, project administration, data curation.

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## Conflicts of Interest

The authors declare no conflicts of interest.

## Data Availability Statement

The data that supports the findings of this study are available in the [Supporting Information](#) of this article.

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## Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** mmi70036-sup-0001-DataS1.pdf. **Table S2:** RNA-seq differential expression for *Borrelia burgdorferi* B31 wild-type and BosR mutant strains under in vitro conditions. **Table S3:** RNA-seq differential expression for *Borrelia burgdorferi* B31 wild-type and BosR mutant strains under mammalian host-adapted (DMC) conditions. **Table S5:** Genome-wide search for BosR binding motifs in *B. burgdorferi* B31. Motifs corresponding to validated BosR sites (*PrpoS\_DR*, *PrpoS\_DR\_rc*, *PospA\_DR1*, *PospA\_DR2*) and the derived AT-rich consensus (ATTTTAAATTAAAT; see Figure S3) were scanned within the 150bp upstream regions of all genes (TSS when available or start codon when no TSS was assigned). Results indicate whether each gene belongs to the RpoS and/or BosR regulons under in vitro and mammalian host-adapted (DMC) conditions.